

1.1 Climate Trajectory and Threshold Exceedance

The spatial distribution of mean annual summer minima and winter maxima across the full dipwell network is presented in Figures 37 and 39 respectively, classified against the Curreli et al. (2013) eco-hydrological thresholds introduced in Section 2. The cluster-level trajectories and threshold exceedance analysis presented in this section quantify the rate and direction of change over the 21-year monitoring period.

1.1.1 Summer and Winter Trajectories

Summer minimum trends were fitted for all five clusters over the monitoring period (Section 3.6.4; Figure 45, panel a). All clusters show declining summer minima, although the rates and significance levels vary substantially. C1 Lake Edge declined at $-0.0096 \text{ m yr}^{-1}$ ($R^2 = 0.196$, $p = 0.052$, $n = 20$) and C5 Coastal Forest at $-0.0377 \text{ m yr}^{-1}$ ($R^2 = 0.396$, $p = 0.003$, $n = 20$) — the only two clusters reaching statistical significance at the 5% level. C2 Dune was marginal ($-0.0122 \text{ m yr}^{-1}$, $p = 0.056$, $n = 21$). C3 Western Residual ($-0.0066 \text{ m yr}^{-1}$, $p = 0.336$) and C4 Main Forest ($-0.0115 \text{ m yr}^{-1}$, $p = 0.398$) were non-significant. The moderate R^2 values across all clusters reflect substantial interannual variability rather than an absence of signal.

These trends translate into progressively worsening threshold compliance. C1 has been below the wet slack summer viability threshold (-0.61 m) in all 20 monitored years, with the shallowest summer minimum recorded in 2016 (-0.704 m). C2 has similarly remained below the threshold throughout the record, with its shallowest minimum in 2005 (-0.682 m). C3 has never approached the wet slack threshold, ranging between -0.955 m and -1.561 m , and falling below the dry slack threshold (-0.98 m) in 20 of 21 years.

Winter maximum trends were fitted for all five clusters. C1 showed a positive but non-significant trend ($+0.0087 \text{ m yr}^{-1}$, $p = 0.119$); C2 ($+0.0062$, $p = 0.477$), C3 ($+0.0034$, $p = 0.769$) and C4 (-0.0034 , $p = 0.850$) were all non-significant. C5 showed a significant declining winter maximum trend ($-0.0368 \text{ m yr}^{-1}$, $p = 0.035$), indicating that this cluster is deteriorating in both seasons.

The C5 Coastal Forest trajectory anomaly warrants specific comment. At -0.038 m yr^{-1} in summer and -0.037 m yr^{-1} in winter, C5's decline rate is 3.9 times the summer rate of C1 — the only other cluster with a statistically significant summer trend — and C5 is the only cluster with significant trends in both seasons. The magnitude of this divergence argues against a purely climate-driven explanation, since all clusters share the same climate forcing. Three candidate explanations, which are not mutually exclusive, are identified: amplified head response per unit climate forcing due to thin saturated thickness at the coastal margin, where a given recharge deficit produces a proportionally larger head change at lower S_y ; progressive coastal boundary retreat, reducing aquifer volume at the seaward edge and

producing an ongoing decline with no equilibrium; and progressive canopy maturation, increasing interception and transpiration demand as the Corsican pine stand ages.

The post-2018 collapse in C5 seasonal amplitude (−18% raw, −21% climate-normalized; Section 4.2.1) — the largest amplitude reduction in the network — reinforces the structural interpretation: recharge peaks are no longer reaching their pre-2018 heights even after correcting for drought summers, suggesting that at least one site-specific process is operating beyond the common climate signal. The non-stabilization of the C5 CUSUM trajectory through to the end of the monitoring record favours an ongoing process (coastal erosion or continued canopy maturation) over a one-off step change.

1.1.2 Network-scale partition of the summer-minimum decline.

The third candidate mechanism for the C5 anomaly — progressive coastal boundary retreat — can be tested quantitatively at network scale. A non-linear panel regression was fitted to the full 21-year monthly record (12,457 observations across 72 wells; FE wells, the scraped well CEH36, and the BACI clearfell-zone wells excluded) to test whether the C5 anomaly is quantitatively consistent with coastal boundary retreat operating site-wide. Each well's perpendicular distance to the eroding Caernarfon Bay High Water Mark was computed within the pipeline from a committed OpenStreetMap mean-high-water coastline (Menai Strait and Llanddwyn Island excluded as non-eroding boundaries). The headline fit is the forest-free panel (C4 and C5 dropped), giving a Dupuit–Forchheimer steady-state strip-aquifer form with a coast-edge anomaly δ_0 of $-29.0 \pm 1.9 \text{ mm yr}^{-1}$ above the climate background, an inland reach L of $894 \pm 52 \text{ m}$ beyond which the gradient effect vanishes, and a climate background c of $-6.4 \pm 0.6 \text{ mm yr}^{-1}$ consistent within one standard error with the network-mean per-well summer-minimum slope. The linear-with-cutoff form and an exponential decay are statistically indistinguishable ($\Delta\text{AIC} = 0.5$); the linear-with-cutoff form is retained on physical grounds, as the Dupuit–Forchheimer steady-state solution for a strip aquifer with a fixed inland reach. The fit is robust to forest-cover exclusion: it is itself the forest-free panel, and the full-network and C3-only fits shift no parameter beyond sampling uncertainty (Script 25; Figure 41).

Applied to C5's mean coast-proximity, the model predicts a gradient contribution of -15.4 mm yr^{-1} , approximately 43% of C5's observed -35.9 mm yr^{-1} summer-minimum decline; the -6.35 mm yr^{-1} climate background accounts for a further 18%. This attribution should be read against the sampling base of the forest-free fit. The forest-free panel that anchors δ_0 and L excludes most C4 and C5 wells, and only one C5 well meets the per-well summer-min slope criterion that feeds the cluster-mean perpendicular distance to coast. The cluster centroid slope itself is computed over all five C5 wells, so the direction of the C5 result is robust — C5 is unambiguously the most coastal-affected cluster — but the precise 41% magnitude is fragile to which C5 well enters the forest-free filter. C1, C2 and C4 all lie beyond the fitted inland reach L ; their observed slopes are accounted for by the climate background plus residual interannual noise. The per-cluster decomposition is tabulated in

Table 16 (and visualized in Figure 41) and itemises each cluster's observed centroid slope partitioned into the climate-uniform background, the coastal-retreat gradient contribution evaluated at the cluster's mean coast-proximity, and the unattributed residual. Progressive coastal boundary retreat therefore moves from a qualitative candidate to a quantified, substantial, but partial contributor to the C5 anomaly. The remaining -14 mm yr^{-1} residual at C5 is consistent with the other two candidate mechanisms named above — amplified head response under thin saturated thickness, and progressive canopy maturation — operating in parallel. Independent corroboration of the BACI ANCOVA's easting $\times$ time covariate against this gradient model is provided in Section 4.6.3 (Table 8).

Cluster	n	Mean dist. to coast	Observed (mm/yr)	Climate (mm/yr)	Coastal (mm/yr)	Residual (mm/yr)
C1 (Lake Edge)	7 / 7	1819 m	-10.9	-6.35	0.00	-4.55
C2 (Dune)	24 / 24	1508 m	-11.2	-6.35	0.00	-4.85
C3 (Western Residual)	19 / 21	826 m	-9.4	-6.35	-2.21	-0.84
C4 (Main Forest)	2 / 9	1142 m	-17.8	-6.35	0.00	-11.45
C5 (Coastal Forest)	1 / 5	419 m	-35.9	-6.35	-15.41	-14.14

Table 1.1.1: Per-cluster decomposition of the observed summer-minimum centroid slope under the forest-free linear-capped panel regression. Climate term is the uniform c parameter representing site-wide background trend; coastal term is $\delta_0 \cdot \max(0, 1 - \bar{d}/L)$ evaluated at the cluster's mean perpendicular distance $\bar{d}$ to the eroding Caernarfon Bay shoreline; residual is the cluster centroid slope minus the climate + coastal model prediction. All slopes in mm yr^{-1} . Negative = declining water table. The residual is negative at every cluster, between -0.8 and -14.1 mm yr^{-1} : the two-term model under-explains the observed decline throughout the network rather than over-fitting it at any cluster, and the shortfall is smallest at C3 Western Residual, the cluster on which the coastal gradient is fitted. (Source: 25_03_cluster_partition.csv).

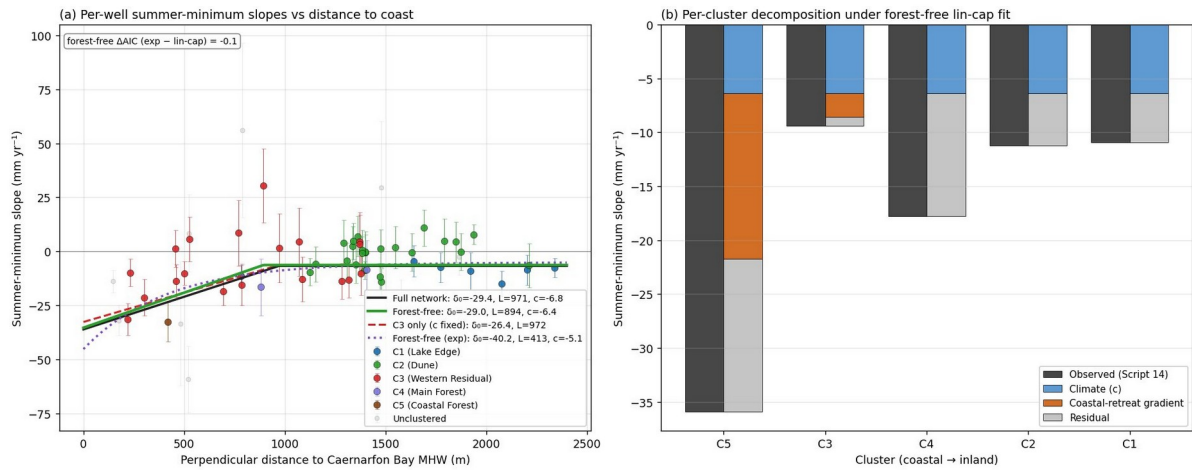


Figure 1.1.1: Network-scale coastal-retreat gradient. Per-well summer-minimum slope (mm yr⁻¹, error bars = ± 1 SE) plotted against perpendicular distance to the Caernarfon Bay High Water Mark, with three linear-with-cutoff fits overlaid: full-network panel (black), forest-free panel after dropping C4 and C5 (green, headline), and C3-only with climate background held to the forest-free value (red, dashed). The three fits are statistically indistinguishable on δ_0 and L . C5 wells (brown) sit closest to the coast; C1, C2 and C4 lie beyond the fitted reach $L = 894$ m. (Source: 25_05_fit_diagnostic.jpg)

1.1.3 Independent transect estimate of the coastal drawdown rate

The δ_0 of the panel regression above is cross-checked directly, and without a model, along a single coast-to-inland transect. Taking MAM spring means from the most coastal well (CEH22, at the dune toe) and an inland anchor beyond the fitted reach L (NW4, ≈ 1 km inland), the coast-minus-inland head difference deepens progressively across 2010–2023 at -28.2 mm yr⁻¹ (95 % CI -34.2 to -22.0 ; $n = 14$; AR(1)-corrected; Figure 42) — within error of the network-scale δ_0 of -29.0 mm yr⁻¹, from an entirely independent construction. The divergence is the coastal well falling — CEH22 declines -21.9 mm yr⁻¹ in absolute terms while the inland anchor is near-flat — rather than the interior rising. This bears on the character of the C5 residual: a static substrate geometry (the thin-saturated-thickness mechanism) would hold the coast-to-inland offset constant, whereas the observed offset grows monotonically (anchor-referenced coastal-end ordering Spearman $\rho = -0.87$, against -0.32 for the raw levels), the signature of a retreating boundary. Spring climate water balance does not trend over the window, so the deepening is not a disguised climate trend. The estimate remains a single line of four wells over a window opening after drawdown was already underway, so it distinguishes a growing (erosion-like) gradient from a static one rather than resolving erosion from a *time-varying* substrate effect (Section 5.7.3).

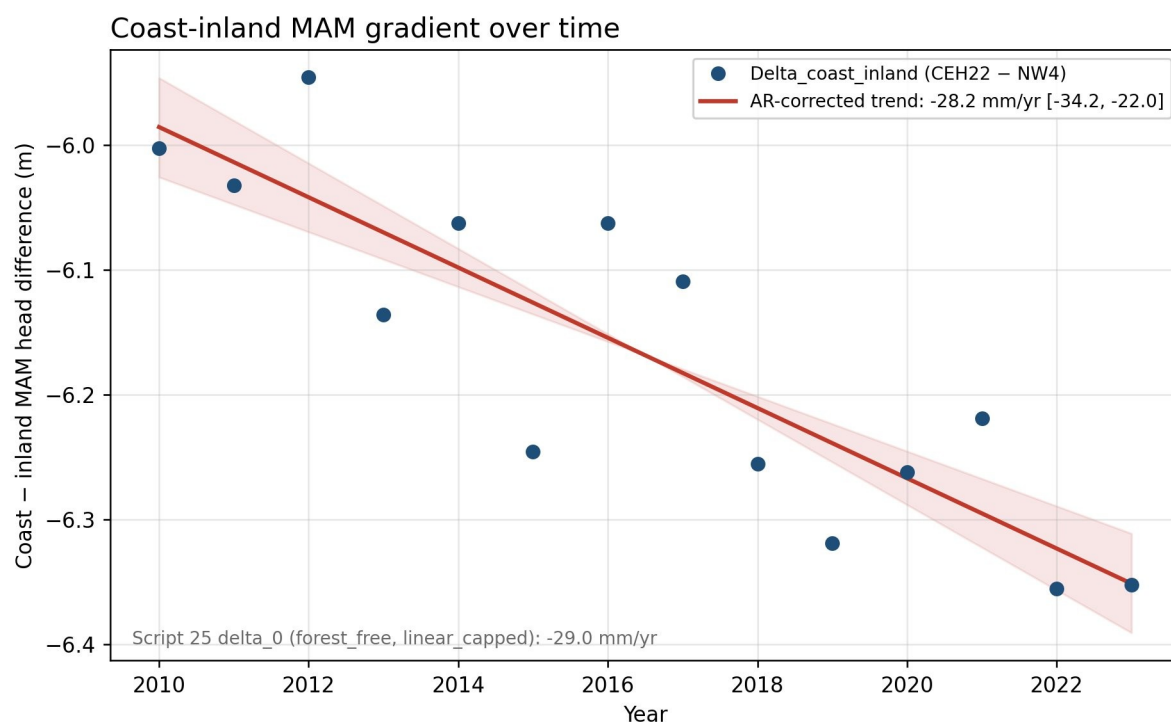


Figure 1.1.2: Independent, model-free estimate of the coastal drawdown rate. The coast-minus-inland MAM head difference (CEH22 at the dune toe minus the inland anchor NW4, ≈ 1 km inland and beyond the fitted reach L) plotted per year over 2010–2023, with the AR(1)-corrected trend (red) and its bootstrap 95% confidence band. The difference deepens at -28.2 mm yr^{-1} (95% CI -34.2 to -22.0 ; $n = 14$), within error of the network-scale δ_0 of -29.0 mm yr^{-1} from the independent gradient regression (grey annotation). A progressively deepening difference is the signature of a retreating coastal boundary; a static substrate geometry would hold it constant. (Source: 38_coast_inland_difference.jpg.)

1.1.4 Observed Winter Maximum Flooding Record

The observed winter flooding record reveals a stark spatial contrast between structural domains (Figure 45, panel b). The C1 Lake Edge cluster exceeded the wet slack winter flooding threshold (-0.10 m) in 14 of 20 monitored years (70%), confirming that the eastern lake-influenced slacks retain substantial winter flooding capacity. C2 Dune exceeded the threshold in 11 of 21 years (52%), indicating episodic but reliable flooding in the eastern mature dune zone. C3 Western Residual exceeded the wet slack threshold in only 1 of 21 years (5%) — occurring in 2007 — and the dry slack threshold (-0.25 m) in 6 of 21 years, confirming that the Western Residual fails to achieve the minimum winter flooding conditions required to maintain even dry slack communities in all but exceptional years. C4 Main Forest did not exceed either threshold in any monitored year. C5 Coastal Forest exceeded the wet slack threshold in 1 of 20 years (5%).

1.1.5 Five-Year Mean Spring Water Level (MSL5)

The cluster-level summer-minimum trajectories examined in Sections 4.8.1–4.8.3 describe stress events — single annual values capturing the deepest water-table excursion of each

year. A vegetation community's response is not driven by stress events alone; it integrates the long-term baseline water level over which it grows. The 5-year mean spring water level (MSL5), as defined in Section 3.7.5 and implemented per van Willegen et al. (2025), is the metric that captures this baseline. It is the rolling 5-year mean of the cluster-mean March–May water table, taken at the end of each hydrological year. Where the summer-minimum metric is sensitive to single dry years, MSL5 is sensitive to sustained hydrological state — the regime within which dune-slack vegetation establishes and persists.

The cluster-mean MSL5 trajectories for window-ends 2014–2025 are shown in Figure 43, and the per-cluster current-window values and threshold-crossing counts are summarized in Table 17. Two horizontal reference values, drawn from Curreli et al. (2013), separate viable hydrological regimes for SD15b wet-slack vegetation (above -0.61 m) and SD16 dry-slack vegetation (above -0.98 m); the SD16 zone is shaded for visual emphasis of cluster–window combinations sitting inside the dry-slack ecohydrological regime.

The lake-edge and open-dune clusters sit comfortably above the wet-slack threshold throughout the recorded window-end range. C1 Lake Edge and C2 Dune both hold their MSL5 within the SD15b regime for every recorded window-end (Table 17). On the baseline metric, the core open-dune system supports SD15b hydrology robustly: the summer-minimum threshold-approach trajectory documented for C1 in Section 4.8.1 reflects a separate stress-event signal that does not extend to the 5-year baseline. The two metrics describe different things and, at the open-dune clusters, point in different directions.

The Western Residual cluster (C3) sits close to but largely above the wet-slack threshold. C3 is below SD15b in six of twelve recorded window-ends (Table 17) and never below SD16, with current-window MSL5 -0.485 m. The cluster's intermittent excursion below SD15b is consistent with its western-margin character (Section 4.9.2) rather than with sustained sub-wet-slack hydrology. C3 is, on the baseline metric, a marginal SD15b cluster: it qualifies as wet-slack viable in approximately half the recorded windows and falls just below it in the others.

The forest clusters present a substantially different picture. C4 Main Forest sits below the SD15b wet-slack threshold for all twelve recorded window-ends, and below the SD16 dry-slack threshold for eight of those twelve. C5 Coastal Forest sits below the SD16 dry-slack threshold for all twelve recorded window-ends — the only cluster to do so. For C4 and C5, the 5-year baseline metric describes a sustained departure from the SD15b vegetation regime, with C5 sustained inside the dry-slack regime as a long-term hydrological state. MSL5 is, in the van Willegen framework, primarily an open-dune metric; the forest clusters' MSL5 deficits are consistent with canopy interception suppressing winter–spring recharge (Section 3.5). The C5 trajectory rests on nine wells (five reference and four extended-network) located in the south-western Corsican-pine block; the cluster is the smallest in the classified network and the result should be read with that in mind, but the wells are spatially coherent and the trajectory is unambiguous within that reach.

The spatial pattern of the MSL5 surface across the network is shown in Figure 44. The figure is an IDW interpolation across the 85 wells that carry at least one valid MSL5 window; for 75 of these the latest valid window ends in 2025, while for eight, recent gaps in spring measurements mean the latest valid window ends earlier (2015–2024), and these appear at their latest available value. The map is distinct from the cluster-mean trajectories of Figure 43 in that it shows the full spatial reach of the per-well MSL5 surface rather than the cluster-identity-based aggregation. The two figures show complementary things — Figure 43 is cluster-mean MSL5 over time, Figure 44 is the per-well MSL5 surface at the most recent window. The map shows the orange-red dry-slack zone concentrated in the south-western corner of the site — the C5 Coastal Forest block — with the surrounding open-dune network in the blue wet-slack range. The Western Residual cluster's western margin appears in the orange-yellow transition zone visible on the map's western edge, consistent with C3's marginal SD15b position in the trajectory analysis.

The most recent window-ends carry a synchronous upward step. All five clusters jump wetter between window-end 2023 and window-end 2024, with the magnitude varying by cluster (Figure 43). The step reflects the 5-year window incorporating the 2023–24 hydrological year, which carried significantly higher annual rainfall than the 2018–22 range it displaces. The magnitude varies because the displacement-sensitive clusters (C4 +220 mm, C5 +132 mm) respond more strongly than the near-surface ones (C1 +50 mm) to a single wet year entering the rolling window. Because the metric is a 5-year rolling mean, the 2024 step's persistence depends on subsequent years: a return to the recent dry hydrological pattern would erode the 2024 jump as it ages out of the window. The 2025 value already shows slight partial reversion at C5 (from -1.162 m at 2024 to -1.140 m at 2025). The 2024 step is therefore a real but climatically-driven recent recovery, not a sustained shift; trend extrapolation from the late-record portion of the trajectory should be avoided.

The MSL5 metric requires a five-year record of valid spring measurements. The earliest valid window-ends fall in 2009–2011 depending on cluster, but the early portion of the record carries only one to five wells contributing per cluster — too few for cluster-mean representativeness, given within-cluster well-to-well variability of order 0.3 m at C2 and 0.4 m at C3 and C4. The trajectory plot starts at window-end 2014, which is the earliest window-end for which all five clusters carry sufficient contributing wells. Earlier window-ends are computed and stored in the supporting outputs for completeness, but are not plotted or discussed here.

The headline ecological reading of the MSL5 trajectory is therefore threefold. The open-dune SAC vegetation regime (C1, C2) is supported by a 5-year baseline well within the SD15b range, and has been throughout the recorded window-end range; this is the central conservation finding for the open-dune system. The Western Residual cluster (C3) is marginal for SD15b — qualifying in some windows and not in others — with the current-window value above the threshold but the longer record showing intermittent excursion

below. The forest clusters (C4, C5) sit in the dry-slack regime as a sustained state, with C5 in particular below SD16 throughout the recorded window-end range; the 2024 recovery does not undo this for C5, and only partially reduces it for C4. Implications for the climate trajectory are taken up in Section 4.10.1; the divergence between this baseline-metric reading and the summer-minimum reading of Section 4.8.1 is returned to in Section 5.1.

Cluster	n wells (2025)	Current MSL5 (m)	below SD15b	below SD16
C1 Lake Edge	8	−0.235	0	0
C2 Dune	26	−0.269	0	0
C3 Western Residual	22	−0.485	6	0
C4 Main Forest	9	−0.739	12	8
C5 Coastal Forest	9	−1.140	12	12

Table 1.1.2: Cluster-mean 5-year mean spring water level (MSL5) at window-end 2025, with counts of window-ends 2014–2025 in which the cluster-mean MSL5 sat below the Curreli et al. (2013) SD15b wet-slack (−0.61 m) and SD16 dry-slack (−0.98 m) thresholds. C5 is the only cluster below SD16 in every recorded window-end. Per-cluster counts are the 74 wells carrying a window-end-2025 MSL5; a further eleven wells have only earlier valid windows and appear in the spatial map (Figure 44) at their latest available value. (Source: 26_msl_5yr_per_cluster.csv).

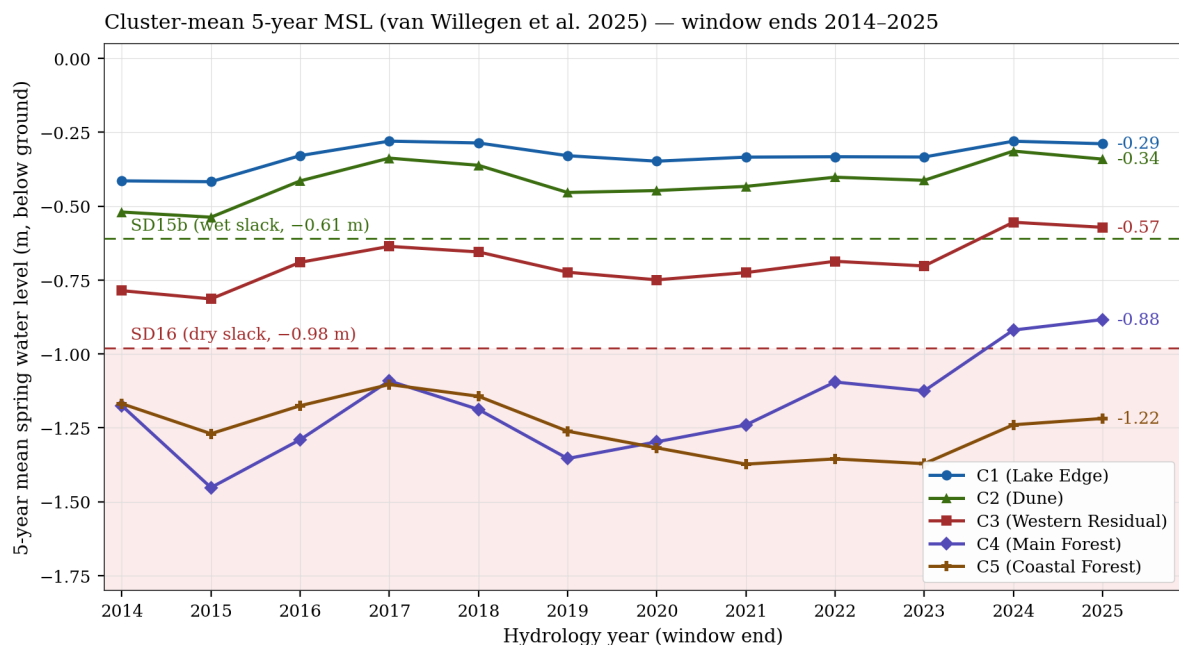


Figure 1.1.3: Cluster-mean 5-year mean spring water level (MSL5) trajectory, window-ends 2014–2025, for the five Pearson-affinity clusters of the Newborough Warren classified network. The dashed horizontal lines mark the Curreli et al. (2013) SD15b wet-slack threshold (−0.61 m) and SD16 dry-slack threshold (−0.98 m); the SD16 zone is shaded for visual emphasis of cluster–window combinations sitting inside the dry-slack ecohydrological regime. C5 Coastal Forest sits below SD16 throughout the recorded window-end range; C4 Main Forest sits below SD15b throughout, and below SD16 for eight of twelve window-ends; C3 Western Residual is intermittently below SD15b; C1 and C2 remain comfortably above SD15b throughout. The

synchronous upward step between 2023 and 2024 reflects the 2023–24 hydrological year (notably wet) entering the rolling 5-year window. Source: Script 26c output fig_msl5_trajectory_report.png).

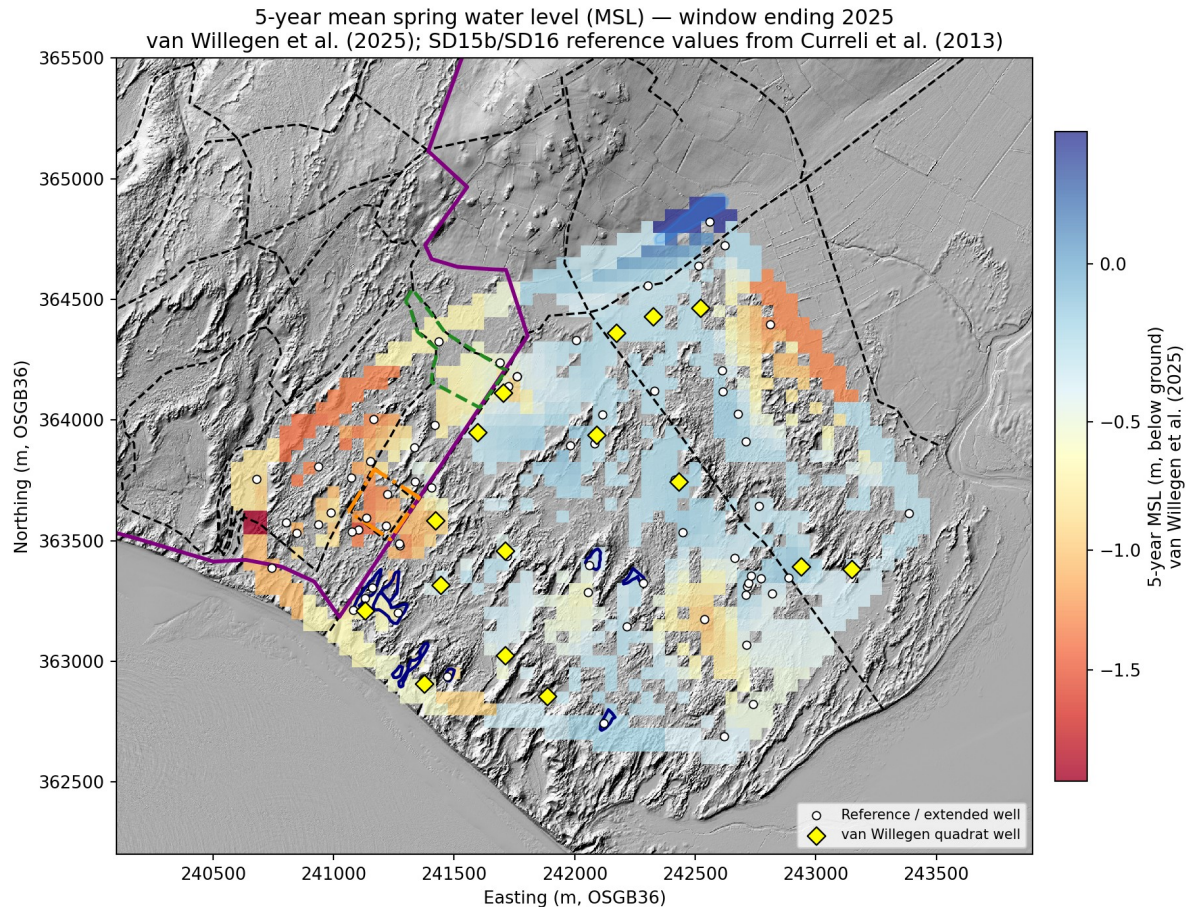


Figure 1.1.4: Spatial pattern of the latest available 5-year mean spring water level (MSL5) per well, IDW-interpolated across the 85 wells carrying at least one valid MSL5 window (of the 88 well classified network; 75 wells at window-end 2025, eight at earlier latest-valid windows; CEH13 and CEH14 excluded for unreliable SSM drainage coefficients). White circles mark reference and extended-network wells; yellow diamonds mark the seventeen van Willegen quadrat-calibrated wells where vegetation community has been empirically calibrated against MSL5. The orange-red zone in the south-western corner of the site corresponds to the C5 Coastal Forest cluster, which sits below the SD16 dry-slack threshold (-0.98 m); the surrounding blue zone across the open dune lies above SD15b. The Western Residual zone visible at the western margin appears in the orange-yellow transition, consistent with C3's marginal SD15b position in the cluster-mean trajectory of Figure 43. (Source: 26_msl_5yr_map.png)

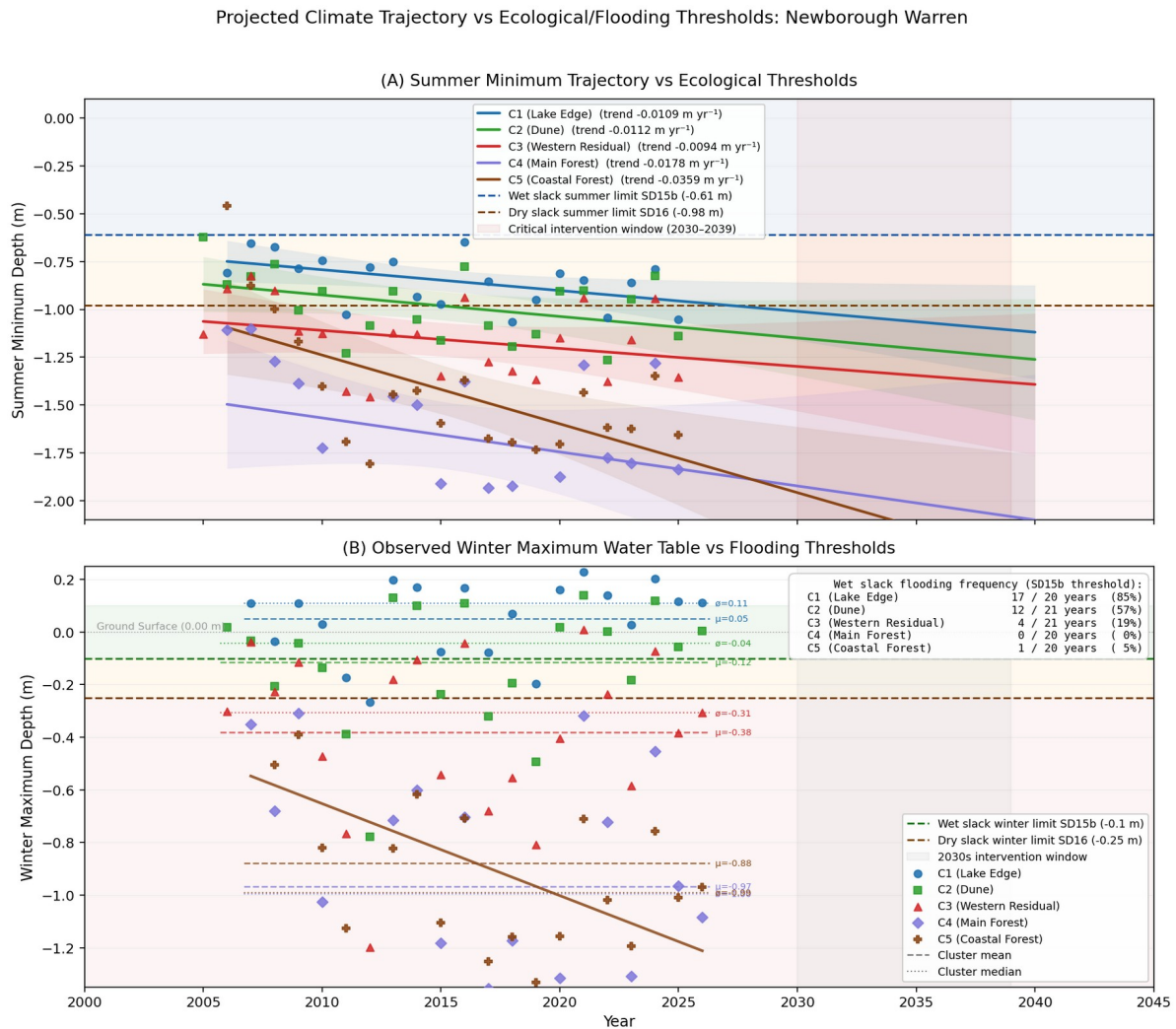


Figure 1.1.5: Climate trajectory and threshold exceedance analysis for Newborough Warren, 2005–2025. Panel (a): observed annual summer minimum water table depth for all five clusters, with OLS linear trend lines and 95% confidence intervals. Dashed horizontal lines indicate the wet slack summer viability limit (SD15b, -0.61 m) and dry slack viability limit (SD16, -0.98 m) from Curreli et al. (2013). Panel (b): observed annual winter maximum water table depth for all five clusters, plotted against the wet slack winter flooding threshold (SD15b, -0.10 m) and dry slack winter flooding threshold (SD16, -0.25 m). The exceedance statistics box reports the proportion of hydrological years in which each cluster exceeded the wet slack flooding threshold over the full monitoring period. (Source: 14_climate_trajectory_stacked.png)

1.1.6 Equilibrium Wetness Index and Vegetation Cross-Validation

MSL5 (Section 4.8.4; defined Section 3.7.5) follows van Willegen et al. (2025), who identified it as the strongest single hydrological correlate of dune-slack vegetation moisture at Newborough. Reconstructing per-well MSL5 for the seventeen van Willegen piezometers reproduces their published mean spring levels to within approximately 11 mm once the fixed dipwell-versus-quadrat datum offset is removed, and between wells MSL5 predicts mean Ellenberg-F — the moisture indicator value (wetness scale 1–12; Hill et al., 1999) — strongly ($r = +0.83$, $n = 18$). The validated wells lie in the lake-edge, open-dune and western-residual clusters, van Willegen having placed no quadrats in the forest interior.

The state-space coefficients afford a structurally-derived estimate of the same wetness. Setting the mean monthly change to zero in the fitted model (Section 3.4) and inverting the head-dependent drainage term yields an equilibrium water-table level, $h_{eq} = (\beta_1 \bar{P} - \beta_2 PE\bar{T})/\beta_3 - d$ ($d = 3.7$ m), from each well's coefficients and the long-term annual-mean climate. Tested directly against Ellenberg-F this equilibrium index performs as well as observed MSL5: the two are statistically indistinguishable as vegetation predictors (Figure 46; Table 18), their correlation confidence intervals overlapping almost entirely and a Williams test for the difference between the two dependent correlations returning $p = 0.81$. The equivalence holds not only in aggregate but well by well: binned by prediction accuracy, the two metrics distribute similarly across match bands, the equilibrium index in fact placing marginally more wells in the closest band (Table 19). That coefficients calibrated purely on month-to-month dynamics predict vegetation moisture as well as the direct spring-level metric is the substantive result, subject to the caveat that eighteen wells give limited statistical power, so this is equivalence within the data rather than proven identity. Because the equilibrium index is computed from the coefficients and climate alone, this comparison is independent of MSL5 — each metric is regressed on Ellenberg-F separately, so the equivalence tests two independent predictors rather than restating the spring-level metric. No single state-space coefficient predicts Ellenberg-F nearly as well (best $|r| \approx 0.5$, versus 0.81 for the equilibrium index), the vegetation signal residing in the equilibrium level rather than any individual rate or storage term.

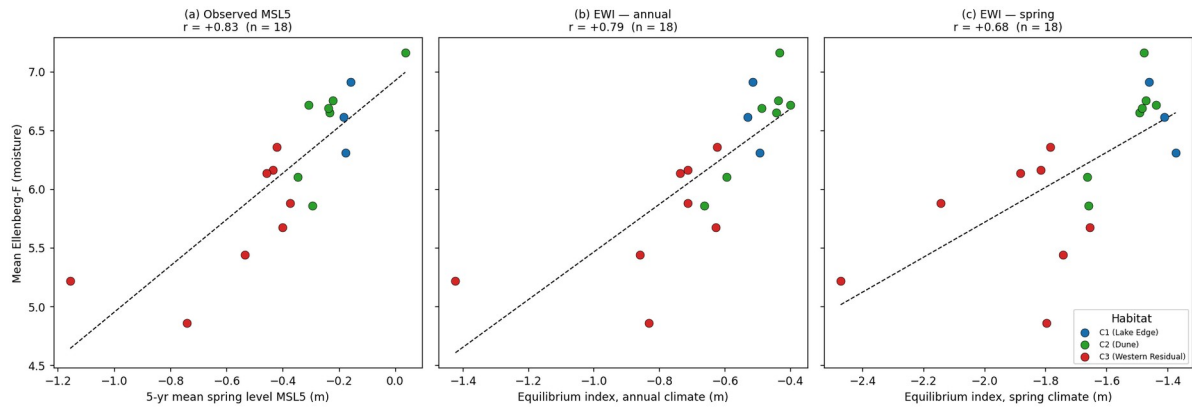


Figure 1.1.6: Between-well prediction of mean Ellenberg-F moisture by three water-table metrics, across the eighteen van Willegen et al. (2024) piezometers carrying a valid state-space fit. (a) Observed five-year mean spring water level (MSL5); (b) equilibrium wetness index on the annual-climate basis; (c) equilibrium wetness index on the spring-climate basis. Points are coloured by hydrogeological cluster (C1 lake-edge, C2 open-dune, C3 western-residual); the dashed line is the ordinary-least-squares fit, with Pearson r and sample size annotated on each panel. The annual-basis index (b, $r = +0.81$) predicts vegetation moisture almost as well as observed MSL5 (a, $r = +0.83$) and more strongly than the spring basis (c, $r = +0.76$), motivating the annual climatology adopted in Section 3.7.6. Ellenberg-F from van Willegen et al. (2024); per-well β from Script 03. Source: 26_ebf_prediction_scatter.png.

	MSL5	Equilibrium wetness index
r [95% CI]	+0.83 [0.59, 0.93]	+0.81 [0.55, 0.93]
RMSE (Ellenberg-F units)	0.337 [0.20, 0.39]	0.352 [0.21, 0.43]

Table 1.1.3: Between-well prediction of mean Ellenberg-F moisture by observed MSL5 and by the equilibrium wetness index, across the seventeen van Willegen et al. (2025) piezometers ($n = 18$ wells). Pearson r and residual root-mean-square error (in Ellenberg-F units) are given with 95% confidence intervals. The two predictors are statistically indistinguishable (Williams' test for dependent correlations, $p = 0.81$; Williams, 1959). Ellenberg-F from van Willegen et al. (2024). Source: 26_ebf_comparison.csv.

Match band	Absolute error (Ellenberg-F units)	MSL5 (wells)	EWI (wells)
A — excellent	≤ 0.15	4	6
B — good	0.15–0.30	7	6
C — moderate	0.30–0.50	5	3
D — poor	> 0.50	2	3

Table 1.1.4: Ellenberg-F prediction accuracy by match band, MSL5 versus the equilibrium wetness index ($n = 18$ wells). Each band counts wells by the absolute prediction error [predicted – observed] in Ellenberg-F units. The two metrics distribute similarly across bands, corroborating the aggregate equivalence of Table 18. Source: 26_ebf_comparison.csv.

Beyond predicting vegetation, the equilibrium index can be calibrated onto the MSL5 scale itself by ordinary least squares. Consistent with MSL5's own open-dune character (Section 4.8.4) and with the forest coefficients being the least constrained on the site (Section 4.9.2), this calibration is scoped to the open-dune network (clusters C1–C3, reference and extended wells): $\text{MSL5} = 0.295 + 0.918 \cdot \text{EWI}$ ($n = 62$, $r = 0.92$, $\text{RMSE} = 119$ mm). Ninety-five per

cent of open-dune wells are reconstructed within 200 mm, and the relationship generalizes beyond the vegetation-calibrated subset, with a root-mean-square error of 100 mm across the 46 open-dune wells outside van Willegen's seventeen-piezometer set against 160 mm on the calibration wells. The C4/C5 forest wells are predicted but held out of scope ($n = 20$, RMSE 227 mm); there the coefficients are least constrained and the estimate is too coarse to place a well across a Curreli threshold. The full per-well reconstruction, with the out-of-scope forest wells flagged, is given in Table S7.1 of the Supplementary Material.

Precision of the two indices

The two indices differ in what limits them. MSL5 averages five spring observations, so its precision is set by how far spring levels swing from year to year: the interannual standard deviation rises from about 175 mm at C1 through 252 mm at C2 and 315 mm at C3 to 277 mm at C5 and 404 mm at C4, giving a five-year window mean a standard error of about 78, 112, 141, 124 and 181 mm respectively. The equilibrium index carries no window, but it divides by β_3 , so its precision is set by how well that coefficient is identified. Propagating each well's β_3 uncertainty gives a standard error of about 315 mm at C1, 452 mm at C2, 511 mm at C3, 429 mm at C5 and 1089 mm at C4.

The equilibrium index is therefore the less precise of the two throughout the network, by a factor of between 3.5 and 6.0 at the cluster medians, and it is more precise than a five-year MSL5 mean at none of the 65 wells where both can be computed. Its advantages lie elsewhere: it requires no five-year spring window, so it can be computed at wells whose spring record is too broken for MSL5, and its precision shows no measurable dependence on record length (β_3 relative standard error against number of observations, Spearman -0.17 , $p = 0.48$ across the extended network). The two indices also degrade at the same wells for opposite reasons — MSL5 where spring levels swing furthest, the equilibrium index where β_3 is weakly identified — and the forest clusters are adverse on both counts at once. Neither should be reported under closed canopy without an explicit precision caveat, and the Ellenberg-F equivalence established above rests on eighteen open-dune piezometers, van Willegen having placed no quadrats in the forest interior.

1.2 Spatial Groundwater Analysis

The per-well SSM reparametrization provides a spatially resolved view of aquifer behaviour that complements the cluster-level summaries of Sections 4.2–4.4. This section presents the spatial coefficient structure across the 88 well classified network (66 reference wells with per-well SSM coefficients; 22 extended wells assigned cluster-level coefficients via the Pearson affinity audit of Section 3.3), the baseline head surface and groundwater flow field, the SSM water balance residual, and equilibrium scenario projections under prescribed climate and forest management perturbations. An interactive scenario viewer (Hollingham,

2026b) integrates these outputs with the Curreli et al. (2013) ecological thresholds and is available as a companion tool to this report.

1.2.1 Datum Sensitivity

Per-well drainage datum optimization (Figure 47a), obtained by independently optimizing the displacement datum D at each well to maximize SSM fit (Section 3.4), shows that the R^2 -maximizing datum varies systematically across the site — from shallow (1.5–2.5 m) at the Lake Edge wells to deep (3.5–5.0 m) in the western and forest interior. However, the R^2 gain from per-well optimization over the uniform 3.7 m value is negligible across most of the network (Figure 47b), confirming the uniform datum assumption adopted in Section 3.4.

Per-well SSM coefficients interpolated across the site (Section 3.4.3) reveal spatially variable patterns whose broad structure aligns with the cluster boundaries (Figure 48a–d), although within-cluster heterogeneity is substantial. All coefficient surfaces are IDW-interpolated to a 50 m grid with dune ridge cells masked where the DEM raster exceeds the interpolated surface by more than 1 m (see Section 3.8).

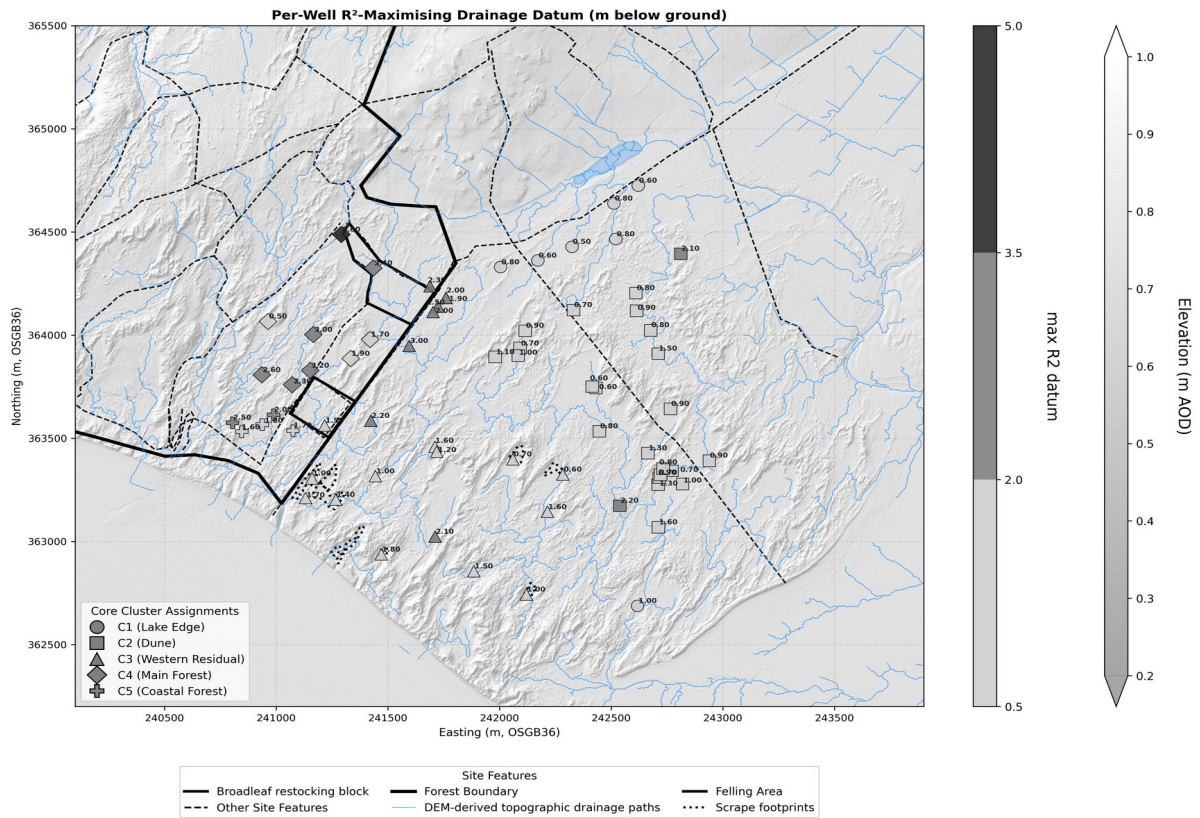


Figure 1.1.7a: Per-well optimal SSM drainage datum across the 66-well reference network, Newborough Warren. Each well's displacement datum D was optimized independently by sweeping from 0.5 to 5.0 m in 0.1 m steps and selecting the value maximizing one-step R^2 (Section 3.4). (a) R^2 -maximizing datum depth (m below ground surface). The east-to-west gradient from shallow (1.5–2.5 m at C1 Lake Edge) to deep (3.5–5.0 m in the western and forest interior) reflects the depth at which the SSM drainage term best captures lateral discharge at each well location. (Source: 03_10_well_datum_r2max_map.png) IDW interpolation (power = 2) over 1 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. LiDAR data: © NRW & OS; see Section 10 for full attribution.

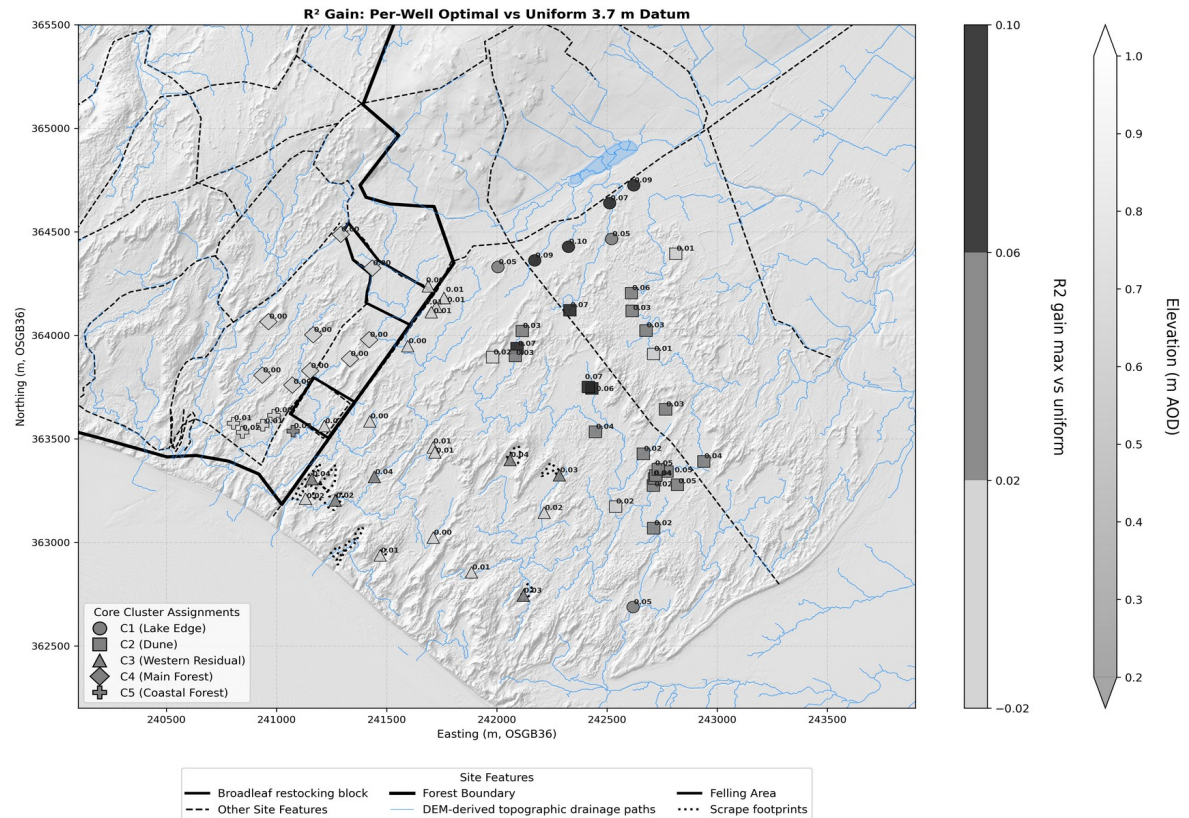


Figure 1.1.7b. Per-well optimal SSM drainage datum across the 66-well reference network, Newborough Warren. Each well's displacement datum D was optimized independently by sweeping from 0.5 to 5.0 m in 0.1 m steps and selecting the value maximizing one-step R^2 (Section 3.4). R^2 gain from per-well optimal datum vs the uniform 3.7 m network datum. Gains are negligible ($<0.01 R^2$) across most of the network, with meaningful improvement (0.02 – $0.04 R^2$) limited to C1 Lake Edge wells where the effective drainage base is shallowest, confirming that the uniform datum assumption does not materially compromise model performance. (Source: 03_10_well_r2_gain_map.png). IDW interpolation (power = 2) over 1 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. LiDAR data: © NRW & OS; see Section 10 for full attribution.

1.2.2 Spatial Coefficient Structure

The R^2 surface (Figure 48a), derived from per-well SSM fits at each well's optimal drainage datum, shows the model explains 70–85% of monthly water table variance across the central and northern open dune, declining toward the forest (C4 and C5 mean 0.68) and some peripheral south-eastern wells. Since the R^2 gain from per-well datum optimization is negligible (Figure 47b), these values also represent the effective fit quality under the standard pipeline datum. The spatial pattern is consistent with the SSM performing best where the climate signal is strongest and least attenuated by canopy or boundary effects.

The three coefficient surfaces (Figure 48b–d) reveal a coherent spatial story about how rainfall enters, is consumed, and drains from the aquifer. Recharge sensitivity (β_1) is highest

in the north-east around the C1 Lake Edge wells closest to Llyn Rhos-Ddu (>5 mm/mm), declines across the open dune (3–5 mm/mm), and drops to 2–3 mm/mm west of the forest boundary. Atmospheric draw (β_2) shows a different geography: the highest values occur in two zones — the C4 Main Forest interior, where canopy-driven transpiration amplifies the signal, and a band across the northern open dune adjacent to the ridge — while C1 Lake Edge wells show notably low values (mean 0.62 mm/mm), consistent with a system where head-dependent drainage dominates the loss pathway. C5 Coastal Forest shows moderate β_2 (mean 1.20), lower than C4 despite also carrying pine canopy, possibly reflecting coastal moderation of atmospheric demand. Drainage rate (β_3 , logarithmic scale) shows the strongest spatial contrast of any coefficient: C1 and northern C2 wells drain fastest ($>8\%$ of displacement per month), the forest interior shows the lowest rates in the network (C4 mean 1.6%, with individual wells below 1%), and C5 falls between the two (mean 4.6%). One C4 well (CEH14) records a negative β_3 (–1.6%), although the estimate is not significantly different from zero ($p = 0.24$), and six of the nine C4 wells likewise carry a non-significant β_3 , so the coefficient is weakly identified across the cluster rather than reversed at this well alone.

Within the C3 Western Residual cluster, the coefficient surfaces resolve into a coherent internal gradient aligned with the south-west to north-east axis — the same axis as distance inland from the eroding south-west coast. Projecting the 21 C3 well positions onto this axis, recharge sensitivity β_1 increases inland ($r = +0.76$, $p < 0.001$), drainage rate β_3 increases inland ($r = +0.58$, $p = 0.009$), and WTF-derived specific yield declines inland, from approximately 0.41 at the south-west margin to 0.25 at the north-east ($r = -0.83$, $p < 0.001$). Because the SSM's β_1 is recharge divided by specific yield, the β_1 and β_3 gradients are both consistent with a single underlying control — a spatial trend in near-surface specific yield — rather than a spatially varying recharge input. Lower specific yield at the inland margin produces both a larger water-table rise per unit recharge and a faster drainage response per unit head. The mechanistic interpretation of this gradient is developed in Section 5.6.1.

The combined picture is of an aquifer whose eastern margin fills quickly, loses water quickly, and resets each season, while its western and forested interior fills slowly, holds water against evaporative demand, and drains sluggishly — the "shallow pan" versus "deep sponge" contrast identified at cluster level in Section 4.2.

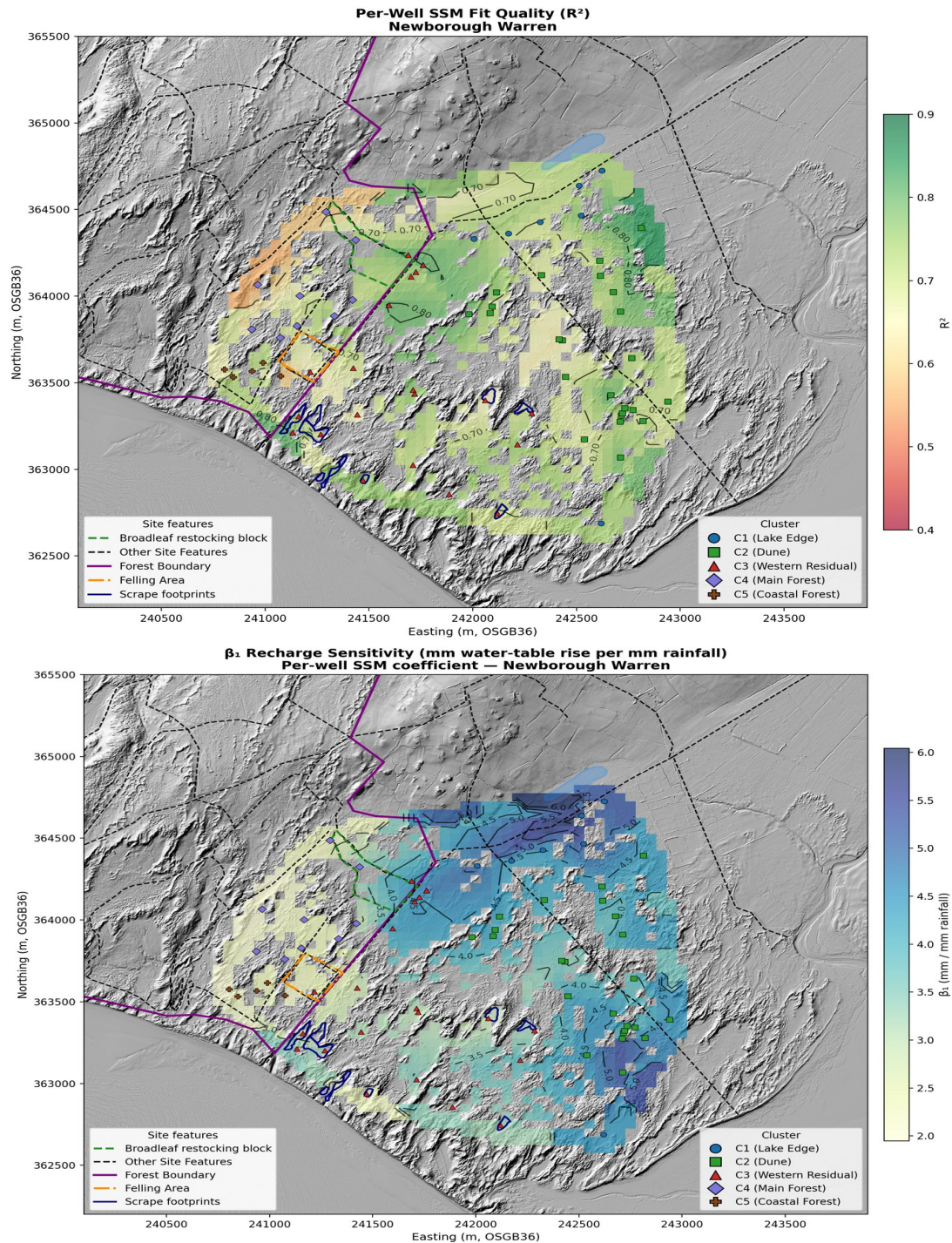


Figure 1.1.8: Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren. Per-well coefficients interpolated using IDW (power = 2) over 1 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. (a) R^2 one-step model fit quality — the SSM explains 65–85% of monthly water table variance across the open dune, declining to 43–72% in the forest interior (C4). (Source: 07_coeff_04_r2_quality.png). (b) β_1 recharge sensitivity (mm water-table rise per mm rainfall) — highest at the Lake Edge (C1, 4–6 mm/mm), with a sharp drop at the forest boundary consistent with 24% canopy interception. Generated by 07_spatial_coefficients.py. (07_coeff_01–04_*.png) LiDAR data: © NRW & OS; see Section 10 for full attribution.

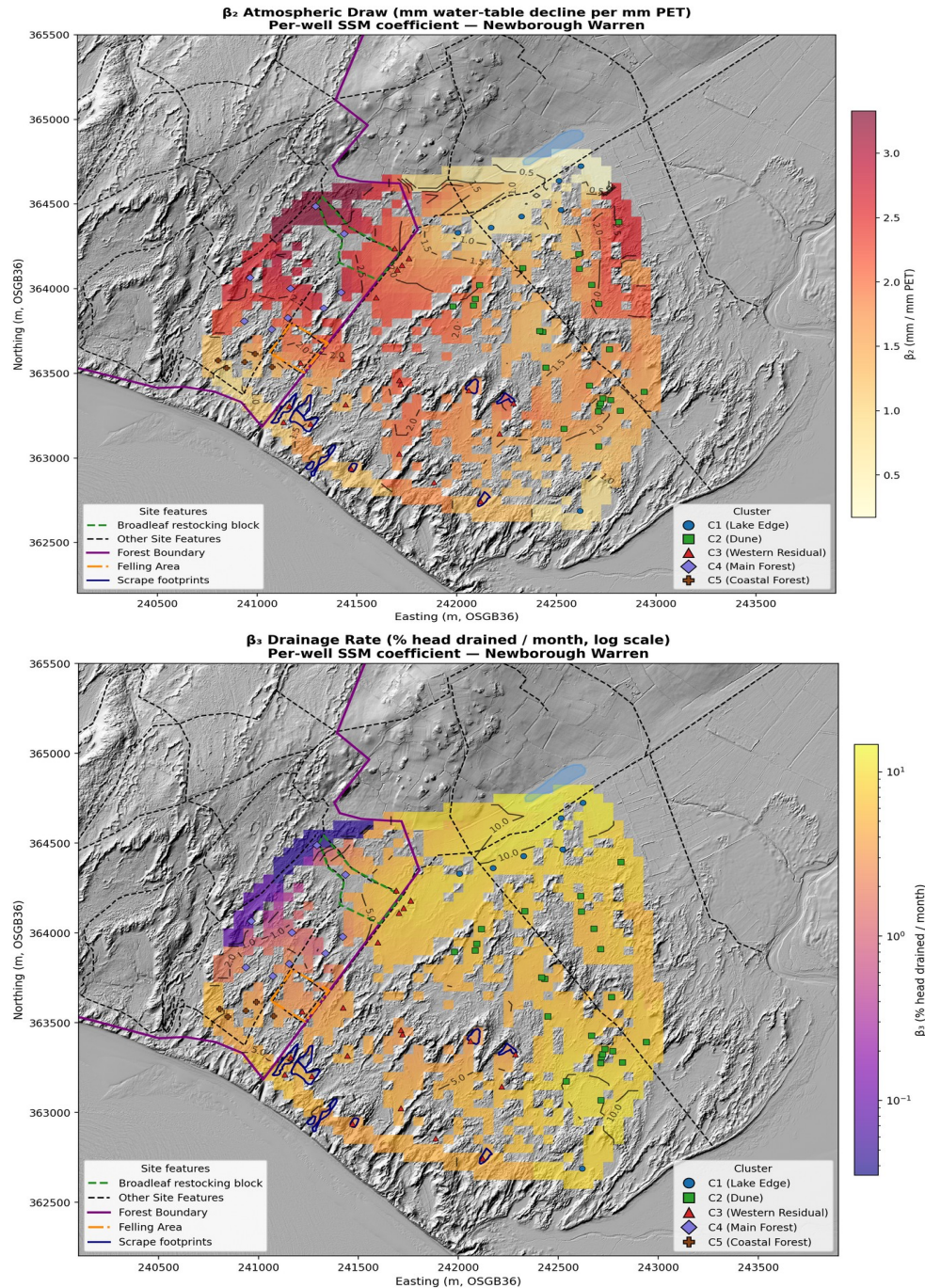


Figure 1.1.8 (continued). Spatial SSM coefficient atlas for the 66-well reference network, Newborough Warren. Per-well coefficients interpolated using IDW (power = 2) over 1 m LiDAR DEM hillshade, with ridge and non-aquifer areas masked. Well markers coloured by cluster assignment. (c) β_2 atmospheric draw (mm water-table decline per mm PET) — highest in the forest interior (C4, 2.0–3.2 mm/mm), lowest at the Lake Edge (C1) and Coastal Forest (C5). (d) β_3 drainage rate (% displacement drained per month, logarithmic scale) — an order-of-magnitude contrast from the fast-draining Lake Edge (7–12% per month) to the near-stagnant forest interior (C4, <3% per month). CEH14, the sole well with negative β_3 , sits at the ridge flank. Generated by 07_spatial_coefficients.py. (07_coeff_01–04_*.png) LiDAR data: © NRW & OS; see Section 10 for full attribution.

1.2.3 Drainage Decay Half-Life and Forest Drawdown Propagation

The drainage decay half-life $t_{1/2} = \ln(2)/\beta_3$ (Figure 49; Section 3.4.6) maps the time for excess groundwater storage above the drainage datum to drain to half its initial value at each reference well. It is derived from the SSM drainage coefficient β_3 alone, without requiring an independent storage estimate. The eastern open dune shows the shortest half-lives in the network: C1 Lake Edge $t_{1/2} = 5.2\text{--}8.0$ months (mean 6.9 months), meaning excess storage is largely dissipated within a single season; C2 Dune $t_{1/2} = 6.2\text{--}12.6$ months (mean 9.9 months). The western dune shows intermediate values: C3 Western Residual $t_{1/2} = 6.2\text{--}22.3$ months (mean 13.8 months), with the wide range reflecting the hydraulic heterogeneity of the western sand body. The forest clusters show markedly longer half-lives: C5 Coastal Forest $t_{1/2} = 12.1\text{--}18.6$ months (mean 15.4 months), comparable to C3 despite carrying pine canopy, reinforcing the elevation and substrate control identified in Section 4.9.5; and C4 Main Forest $t_{1/2} = 16.1\text{--}88.1$ months (mean 39.8 months), the longest half-lives in the network. At the upper end of the C4 range, excess groundwater storage persists for over seven years before draining to half its initial level. The pronounced C4 hotspot on the elevated plateau, where β_3 is lowest, reflects the progressive thinning of the sandy aquifer over the underlying bedrock ridge (Section 2); its detailed geometry beneath C4 remains unconstrained without coring. The wide within-C4 range (16–88 months) is itself a finding, indicating that the forest block is hydraulically non-uniform. CEH14 (negative β_3) and CEH13 (near-zero β_3) are excluded as ridge-flank wells where β_3 is not reliably identified; $t_{1/2} = \ln(2)/\beta_3$ cannot meaningfully be formed from a negative or near-zero coefficient.

The model benchmarking diagnostic (Section 4.4; Figure 16) complements the half-life map. The iterative NSE improvement from SSM over TLM (ΔNSE) correlates positively with β_3 ($r = 0.68$, $p < 0.001$) and negatively with β_2 ($r = -0.67$, $p < 0.001$). The TLM fails most severely at C1 Lake Edge (median $\Delta\text{NSE} = +1.29$, median TLM NSE = -0.57), where head-dependent drainage to the lake is strongest and a fixed drainage rate cannot capture the large seasonal variation in discharge. Conversely, the TLM performs relatively well at C4 Main Forest (median $\Delta\text{NSE} = +0.23$, median TLM NSE = $+0.42$), where β_3 is so weak that misspecifying the drainage term has little impact on overall model performance. ΔNSE therefore identifies where drainage memory is indispensable - the eastern, lake-proximal aquifer - and where atmospheric draw dominates the water balance regardless of drainage formulation - the forest interior.

When $t_{1/2}$ and ΔNSE are plotted together with point size scaled by S_y (Figure 51), the five cluster centroids occupy distinct regions of the diagnostic space, though the separation is moderate and individual wells overlap appreciably within it. C1 Lake Edge (short $t_{1/2}$, high ΔNSE , low S_y) represents a shallow, fast-cycling aquifer over flat clay/till with a short, highly head-dependent drainage path to the lake. C4 Main Forest (long $t_{1/2}$, low ΔNSE , intermediate S_y) represents a sluggish system in which β_3 is too weak for its misspecification to matter - consistent with drainage impeded by the irregular bedrock base the forest is known to overlie. C5 Coastal Forest is displaced from C4 in $t_{1/2}$ and storage despite sharing the same pine canopy - occupying intermediate $t_{1/2}$ and higher S_y , consistent with deeper, more uniform sand at its lower elevation; its higher ΔNSE marks where the state-space model more clearly outperforms the linear benchmark. C2 and C3 span the middle ground,

with C3's longer $t_{1/2}$ and higher S_y reflecting the deeper western sand body. The separation of clusters in this independently derived diagnostic space provides a model-based characterization of aquifer architecture that is consistent with, but independent of, the Ward's clustering applied to the raw hydrograph data. This is the per-well quantification of the 'shallow pan' versus 'deep sponge' contrast identified at cluster level in Section 4.2.

The spatial reach of forest management perturbations is shown directly by the drawdown propagation analysis (Figure 50), which models the steady-state head perturbation from forest canopy interception as an exponential decay with flow-weighted cost-distance from the forest boundary ($\lambda \approx 230$ m). Estimated drawdown exceeds 50 mm only within ~ 100 m of the forest edge, decays to 24 mm at CEH23 and 8 mm at CEH6, and is negligible (≤ 1 mm) at all C1 and eastern C2 wells. These are steady-state values; the timescale to reach equilibrium is governed by the drainage decay half-life values mapped in Figure 49, ranging from months in the eastern dune ($t_{1/2}$ approximately 7-14 months) to years in the forest interior (C4 mean $t_{1/2}$ approximately 40 months) (Section 5.4.3). Forest management perturbations are therefore structurally confined to the forest zone and its immediate western margin; they do not propagate to the eastern slacks where the ecological need is greatest.

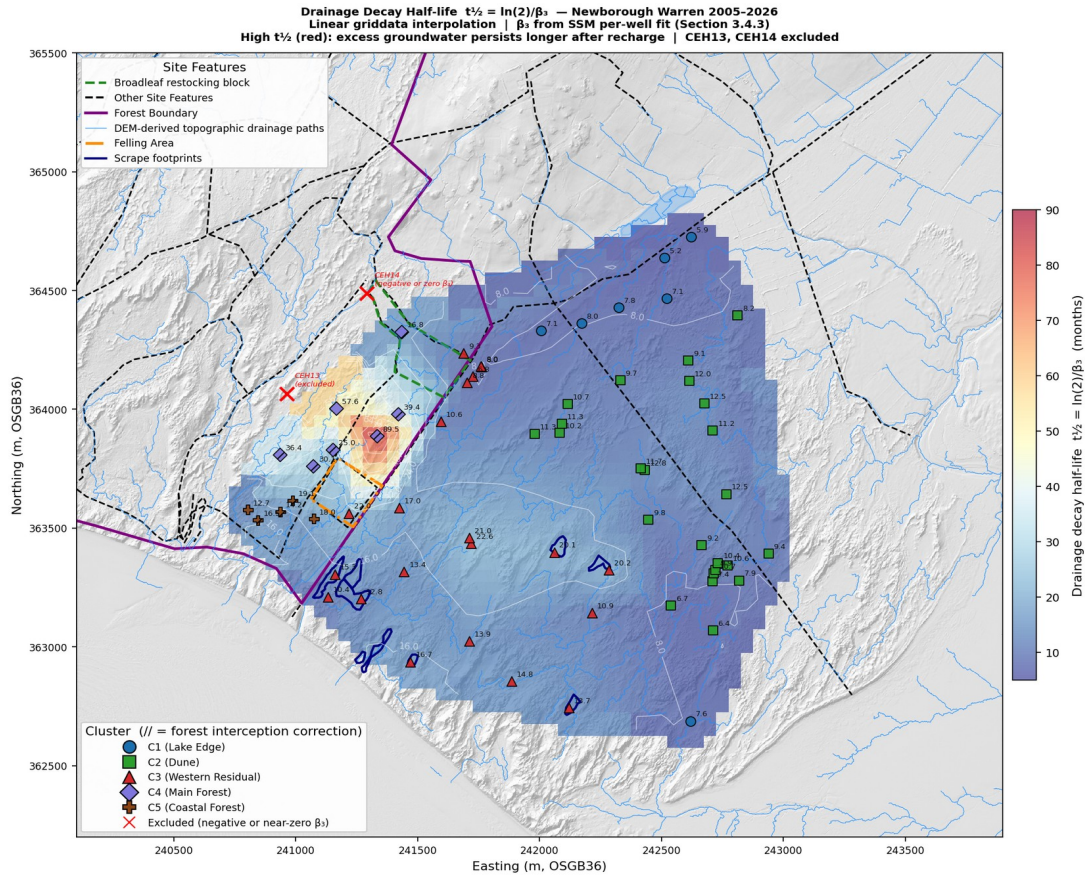


Figure 1.1.9: Drainage decay half-life ($t_{1/2} = \ln(2)/\beta_3$, months) across the 66-well reference network (CEH 13 and 14 excluded), Newborough Warren 2005-2026. The half-life is the time for excess groundwater storage above the drainage datum (3.7 m below ground level; Section 3.4.1) to drain to half its initial value through natural discharge alone, derived from the per-well SSM drainage coefficient β_3 (Section 3.4.3). Low values (blue) indicate rapid aquifer turnover; high values (red) indicate persistent storage following recharge. For formal comparisons, the mathematically equivalent recession e-folding time is $t_R = 1/\beta_3 = t_{1/2} / \ln(2)$. The eastern open dune contrasts sharply with the forest interior: C1 Lake Edge ($t_{1/2} = 5$ -8 months) drains within a season; C4 Main Forest ($t_{1/2} = 16$ -88 months, mean 40 months) retains excess storage for months to years, with a pronounced hotspot on the elevated plateau where β_3 is lowest. The wide within-C4 range reflects the progressive thinning of the sandy aquifer over the underlying bedrock ridge. C5 Coastal Forest ($t_{1/2} = 12$ -19 months) resembles C3 Western Residual ($t_{1/2} = 6$ -22 months) despite carrying pine canopy, consistent with an elevation and substrate control (Section 4.9.5). Two wells are excluded: CEH14 (negative β_3 , $t_{1/2}$ undefined) and CEH13 (near-zero β_3 , $t_{1/2}$ approximately 526 months), both ridge-flank settings where the displacement formulation does not apply; shown as red crosses. IDW interpolation (power = 2) over 1 m LiDAR DEM hillshade, site boundary clipped. Well markers coloured by cluster; hatched markers indicate forest interception correction applied to Sy. LiDAR data: NRW and OS; see Section 10 for full attribution. (Source: 18_wtf_05_half-life_map.png)

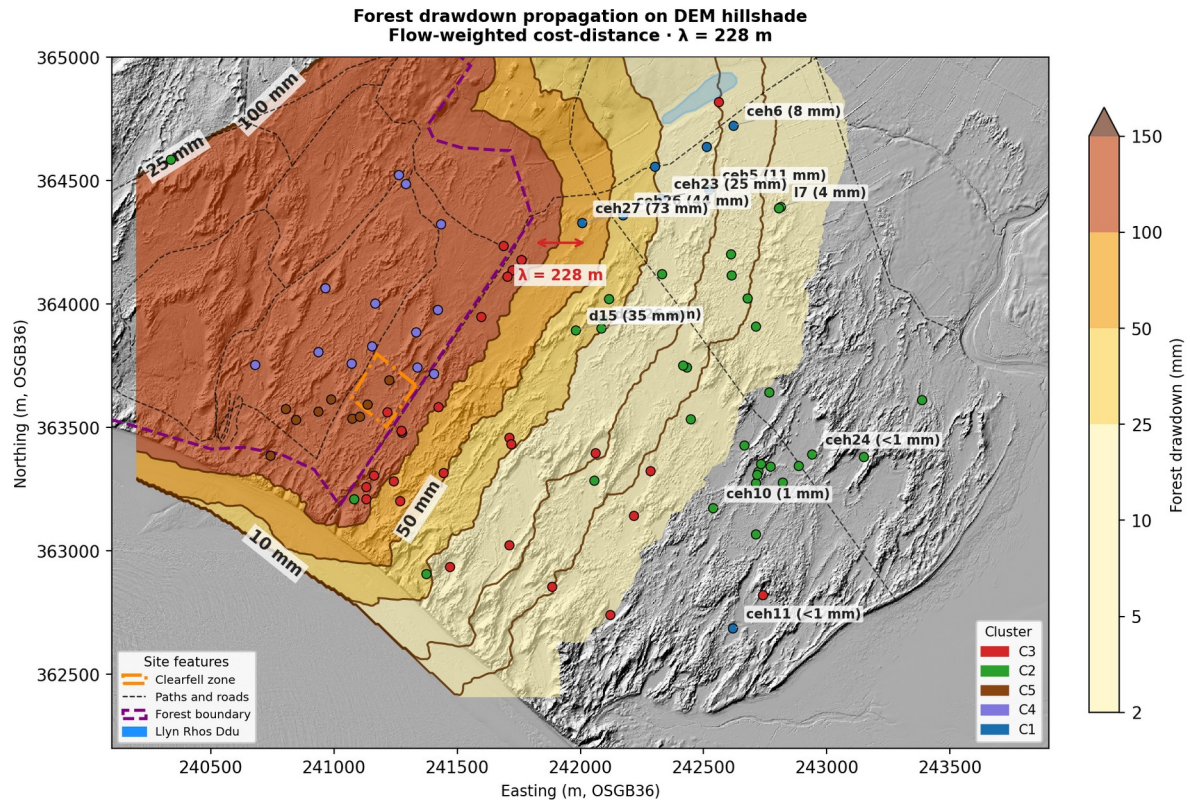


Figure 1.1.10: Forest drawdown propagation across the Newborough Warren dune aquifer, showing the estimated steady-state head perturbation (mm) attributable to Corsican pine canopy interception. Drawdown at each well is modelled as an exponential decay with flow-weighted cost-distance from the forest boundary, using a characteristic length scale $\lambda \approx 230$ m. The background surface shows mean annual water table elevation (m AOD) with normalized Darcy flow direction vectors (white arrows) derived from the IDW head gradient field. Labelled wells show estimated drawdown in parentheses: the signal exceeds 50 mm only within ~ 100 m of the forest edge (D15: 34 mm), decays to 25 mm at CEH23 and 8 mm at CEH6, and is negligible (≤ 2 mm) at all C1 Lake Edge and eastern C2 Dune wells (CEH24, CEH10, CEH11). The 10 mm contour approximately coincides with the 100 m cost-distance ring from the forest boundary. Forest management perturbations are structurally confined to the forest zone and its immediate western margin. Well markers coloured by cluster. Background: mean annual water table surface (m AOD; Figure 57, Section 4.9.6) with normalized Darcy flow direction vectors from the IDW head gradient field. LiDAR data: © NRW & OS; see Section 10 for full attribution. (Source: 20_drawdown_propagation_nohead.png).

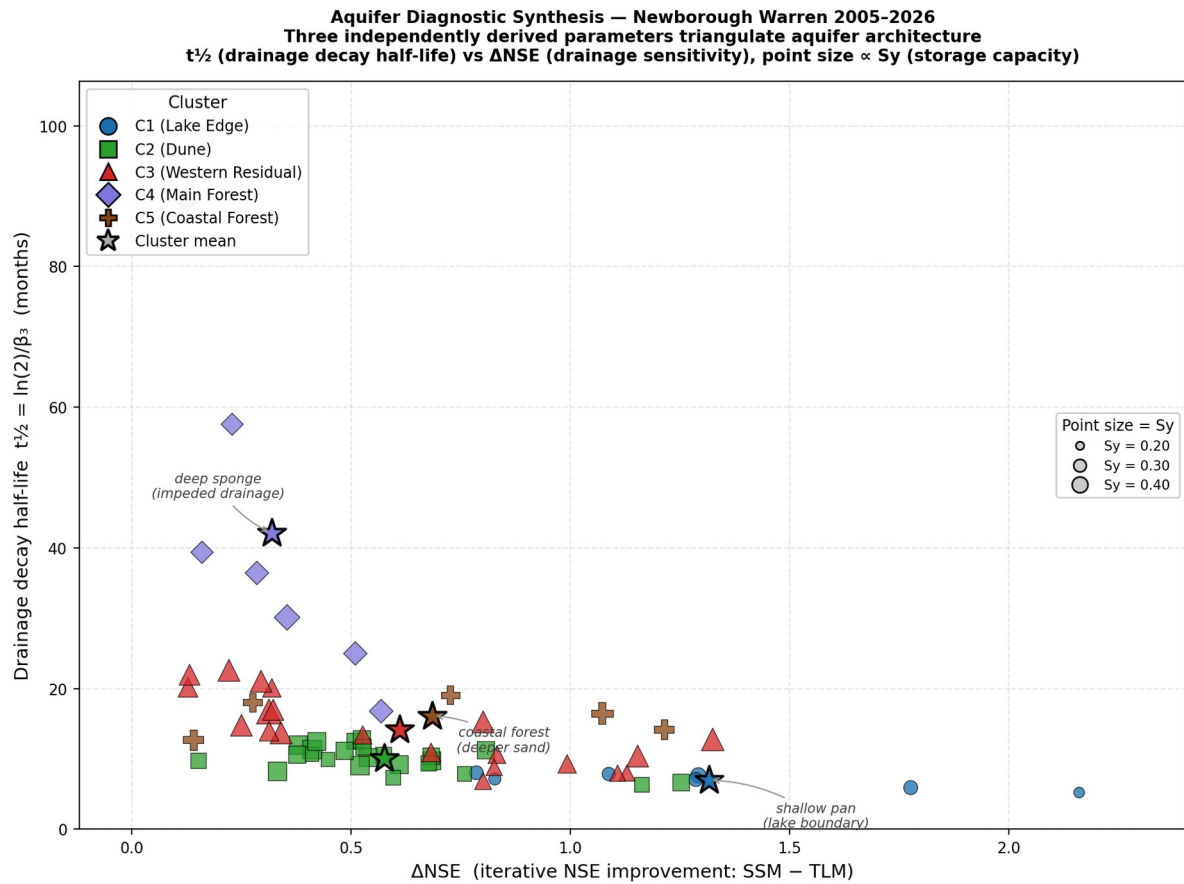


Figure 1.1.11: Aquifer diagnostic synthesis across the reference well network, Newborough Warren 2005–2026. Drainage decay half-life ($t_{1/2} = \ln(2)/\beta_3$, months) plotted against iterative NSE improvement ($\Delta NSE = NSE_{SSM} - NSE_{TLM}$), with points coloured by cluster and sized by WTF-derived specific yield (Sy). Large starred markers show cluster means. The five cluster centroids occupy distinct regions of the diagnostic space, though the separation is moderate and individual wells overlap appreciably within it: C1 Lake Edge (short $t_{1/2}$, high ΔNSE) represents a shallow, fast-cycling aquifer where head-dependent drainage is critical for prediction; C4 Main Forest (long $t_{1/2}$, low ΔNSE) represents a sluggish system where atmospheric draw dominates and drainage is weak, consistent with drainage impeded by the irregular bedrock base the forest is known to overlie. C5 Coastal Forest separates from C4 on both axes despite sharing pine canopy, consistent with the elevation contrast and the substrate inferred to accompany it (Section 4.9.5). The three parameters — $t_{1/2}$ (drainage decay half-life), ΔNSE (drainage sensitivity), and Sy (storage capacity) — are estimated by distinct methods, yet together characterize aquifer architecture without requiring borehole data. CEH14 (negative β_3) and CEH13 (near-zero β_3) excluded. (Source: 18_wtf_06_aquifer_diagnostic_synthesis.png).

1.2.4 Coastal-Margin Processes and the Easting Gradient

The forest drawdown propagation map (Figure 50) raises a natural comparison: how does the spatial reach of canopy-driven drawdown compare with the coastal-margin processes acting on the same aquifer? Two such processes operate at the western boundary, and because they act in opposing directions, neither can be assessed in isolation. Shoreline retreat lowers the ground surface relative to the water table, raising the displacement head and enhancing seaward drainage — the same mechanism, in reverse sign of intent, as the dune scraping of Section 4.5. Sea-level rise acts oppositely, raising the water table at the coastal boundary. Mapped as companion fields on the same hillshade base as the forest drawdown figure, constructed as set out in Section 3.4.5, these processes are shown in Figure 52 (erosion drawdown from a single shoreline-retreat event), Figure 53 (the sea-level-rise head response as a finite-window transient), and their cell-by-cell difference in Figure 54.

The net map (Figure 54) shows that the two processes do not cancel but partition spatially. A narrow erosion-dominated band lies along the dune-toe margin, coincident with the western Coastal-Forest (C5) and western Western-Residual (C3) wells; this is flanked by a sea-level-rise-dominated rise at the immediate shore and near-neutral conditions across the dune interior. The erosion and sea-level-rise contributions are of comparable magnitude over a matched five-year window, so the western-margin signal is the residual of two opposing processes rather than the action of either alone.

This spatial partition provides a physical interpretation of the easting $\times$ time covariate carried in the clearfell BACI analysis (Section 4.6.3). That covariate absorbs a west-to-east deepening gradient that the BACI design cannot otherwise separate from the felling signal. The forest-free coastal-retreat gradient fitted in the corroboration analysis predicts a western-margin drawdown of -11.1 mm yr^{-1} , which is statistically consistent with the $-16.8 \pm 5.0 \text{ mm yr}^{-1}$ absorbed by the easting $\times$ time term at the Forest-Impact tier ($z = -1.16$; Section 4.6, Table 8). A single linear easting term is therefore a defensible first-order proxy for the western-margin coastal signal, even though that signal in fact combines processes of opposing sign and differing inland reach: erosion acting at the dune toe, sea-level rise at the shore. The covariate captures the net west-to-east gradient but does not decompose it into its constituent processes; the interpretive consequences for the clearfell attribution are taken up in Section 5.5.

These three figures are single-mechanism illustrative constructions on the eroding Caernarfon Bay shore only; they combine a single storm-magnitude retreat event with gradual sea-level rise, and depend on assumed retreat-rate, hydraulic-diffusivity, datum, and window parameters. They illustrate the relative spatial reach of the two processes, not a deterministic forecast or a closed water balance.

The dune scraping of Section 4.5 can be placed on the same spatial footing as the forest and coastal fields. Figure 55 renders the April 2015 CEH36 scrape as a steady-state drain, with edge magnitude anchored to the measured CEH36 response (129 mm) and the same leaky-

aquifer decay length as the forest field ($\lambda \approx 230$ m). On this anchoring the unsurveyed excavation depth is recovered as an inferred ≈ 0.42 m. Modelled drawdown exceeds 50 mm only within ~ 100 m of the excavation and decays to a few millimetres by the nearest clearfell-tier wells. As with the forest field these are steady-state values: the equilibration time at these distances is years to decades (Section 5.4.3), which is why the uphill propagation is not resolved as a distance-decay signal within the 31-month BACI window (Section 4.5).

Figure 56 combines the two non-felling drawdown drivers acting on the clearfell compartment — the 2015 scrape and a Storm-Brendan-scale retreat event — as a single head-loss field on the pre-fell baseline. The combined drawdown reaches ≈ 135 mm at the scrape and tapers through the clearfell compartment, with the coastal-erosion term adding a south-western band; the felling step estimated by the BACI ANCOVA (Section 4.6) is the residual once both are accounted for. This is the spatial form of the easting $\times$ time covariate of Section 4.6.3: the covariate absorbs the net west-to-east gradient that this figure decomposes into the scrape drain and the coastal retreat.

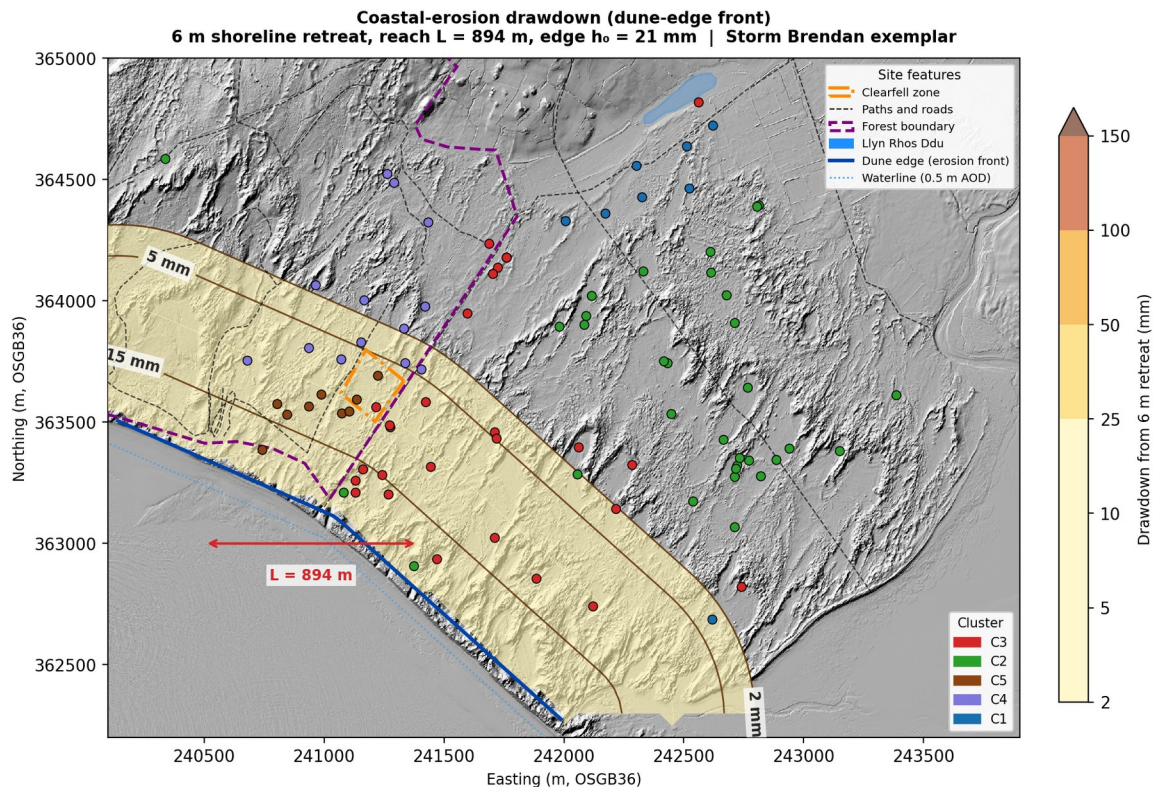


Figure 1.1.12: Coastal-erosion drawdown at the western dune margin, Newborough Warren, on the DEM hillshade base. The field shows the water-table drawdown produced by a single shoreline-retreat event, decaying inland from the dune-toe front (the DEM mean-waterline contour offset 100 m landward) following the forest-free linear-capped coastal-retreat gradient fitted in Section 4.6 (Table 8). Edge magnitude is set by the long-term retreat rate; δ_0 and the characteristic length L are read live from the Script 25 fit. The field is clipped to the study-area outline and references the eroding Caernarfon Bay (south-western) shore only. (Source: 20_coastal_erosion.png).

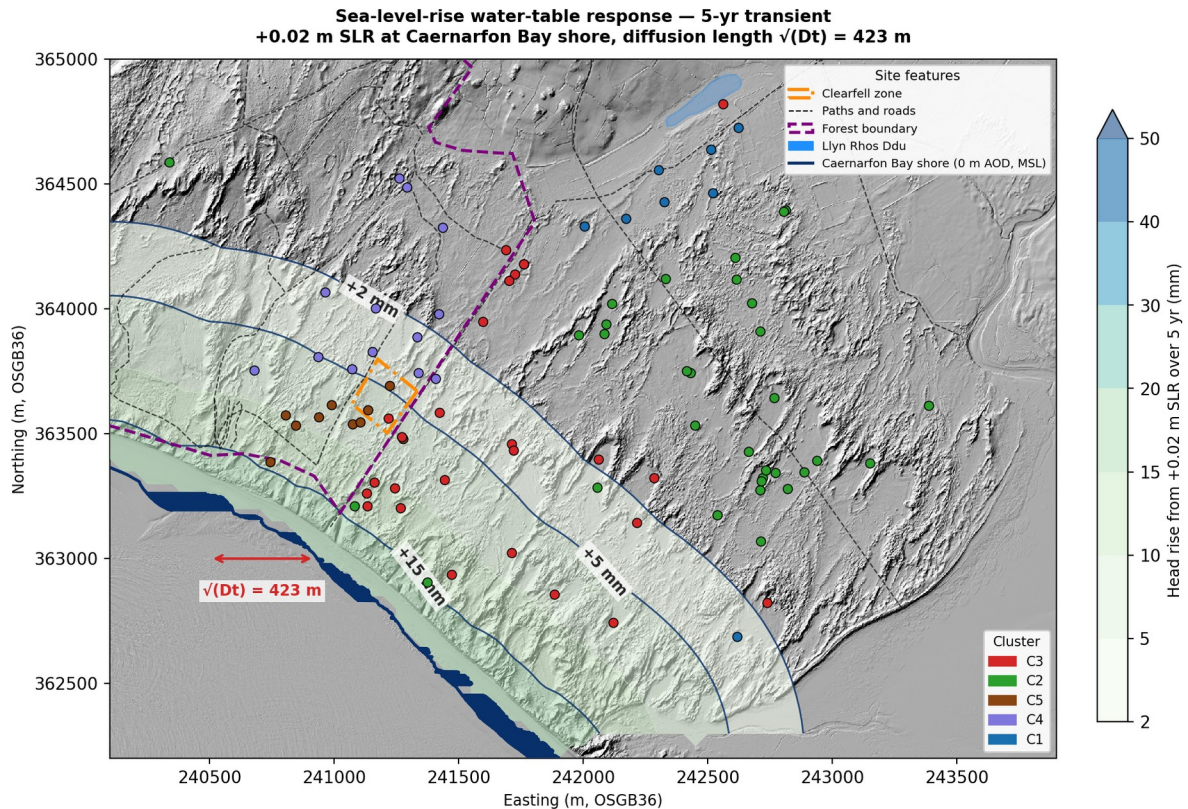


Figure 1.1.13: Sea-level-rise head response at the western dune margin, on the same hillshade base and clip as Figure 52. The field shows the water-table rise produced by gradual sea-level rise over a five-year window, modelled as a boundary perturbation diffusing inland, $\Delta h(d) = \Delta SLR \cdot \text{erfc}(d / (2\sqrt{(Dt)}))$, with hydraulic diffusivity $D = Kb/Sy$ (K and b from literature; Sy the C3-cluster median from the water-table-fluctuation analysis, Section 3.7.3). The boundary is referenced to mean sea level (0 m AOD), distinct from the dune-toe front of the erosion field. (Source: 20_slr_response.png).

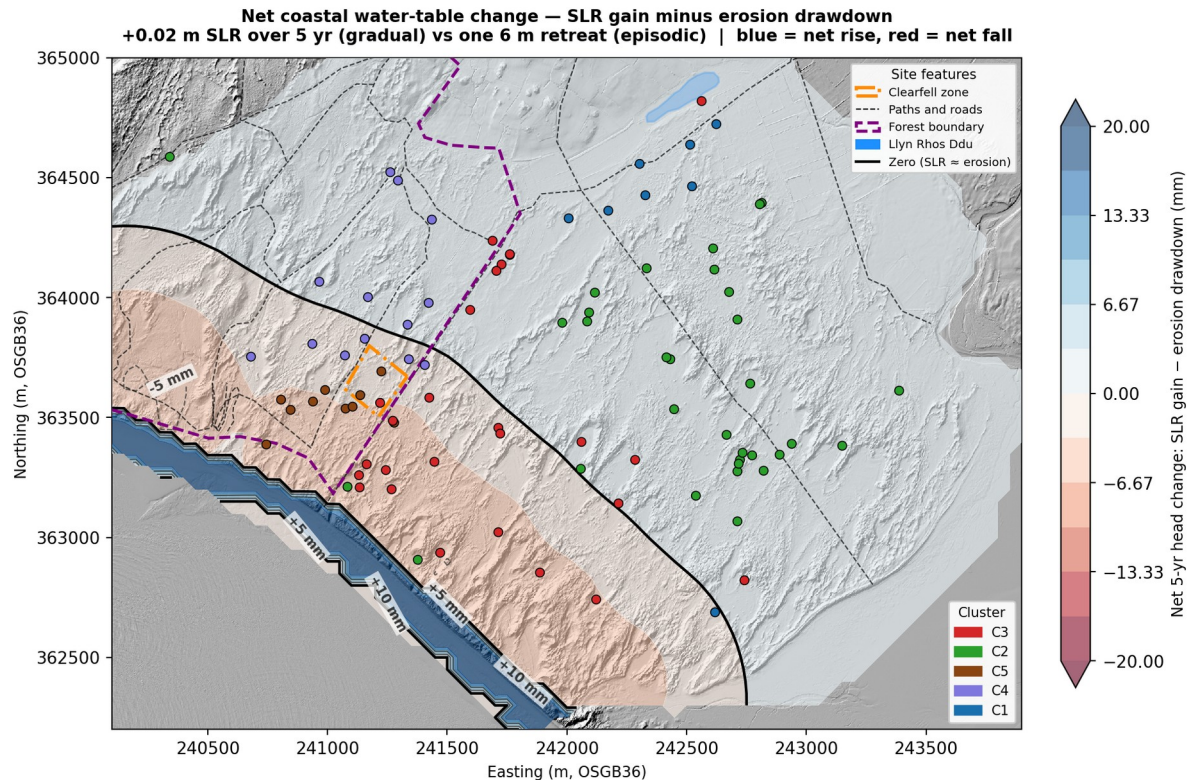


Figure 1.1.14: Net coastal-margin change at the western dune margin: the cell-by-cell difference between the sea-level-rise head gain (Figure 53) and the erosion drawdown (Figure 52) over a matched five-year window, on a diverging scale with the zero contour emphasized. The two processes partition spatially rather than cancelling: a narrow erosion-dominated band (net deepening) along the dune-toe margin, coincident with the western Coastal-Forest (C5) and Western-Residual (C3) wells, flanked by a sea-level-rise-dominated rise at the immediate shore and near-neutral conditions across the dune interior. This spatial partition provides a physical interpretation of the easting $\times$ time covariate in the clearfell BACI analysis (Sections 4.6, 4.9.4). The figure is a single-mechanism illustrative construction on the Caernarfon Bay shore only, combining an episodic retreat event with gradual sea-level rise under assumed retreat-rate, diffusivity, datum, and window parameters; it depicts relative spatial reach, not a deterministic forecast or a closed water balance. (Source: Script 20 output 20_coastal_net_effect.png).

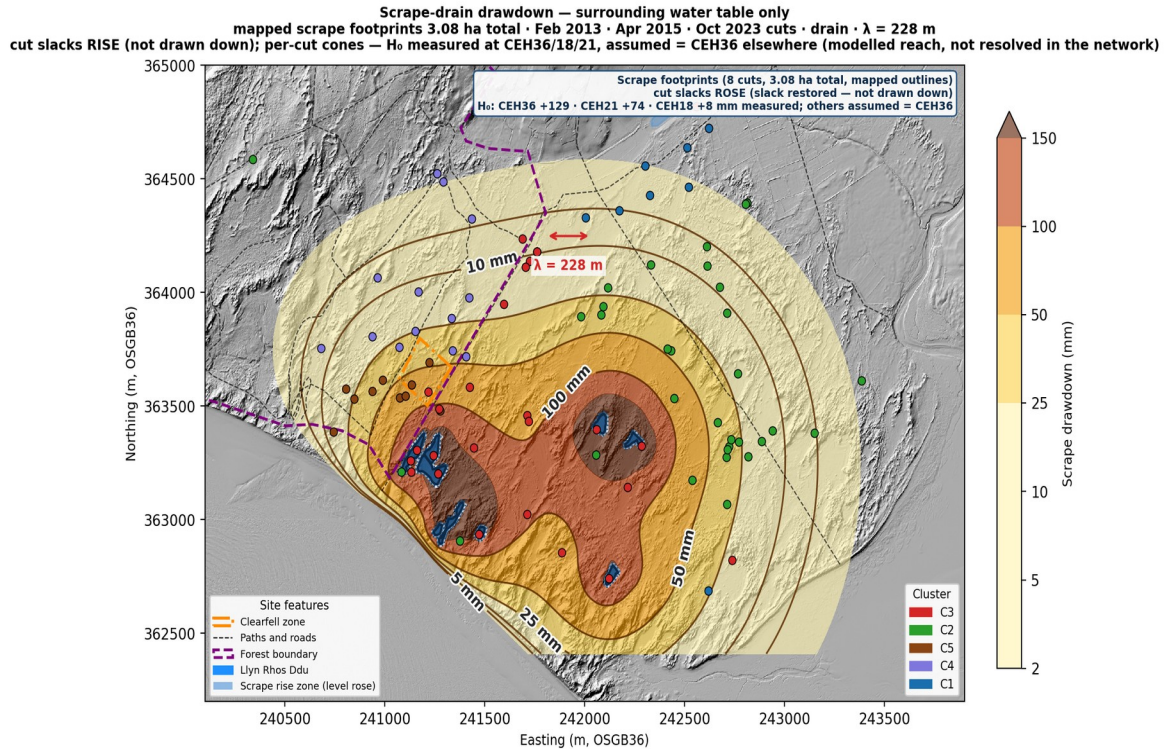


Figure 1.1.15: Dune-scrape drawdown across all eight mapped cuts, Newborough Warren (Feb 2013, Apr 2015, Oct 2023), on the DEM hillshade base. All cuts are treated as permanent co-active features; the displayed field is the long-term equilibrium superposition $\sum_i H_{0,i} \cdot [\exp(-d_i/\lambda) - \exp(-d'_i/\lambda)]$ where d_i is Euclidean distance to cut i 's footprint, d'_i is distance to its mirror image reflected across the OS High Water Mark coastline (method of images; enforces zero drawdown at the fixed-head sea boundary), and $\lambda \approx 230$ m is the leaky-aquifer decay length (live C3 Sy and β_3 ; $K=6$ m d $^{-1}$, Betson & Bristow 2002; $b=5$ m). Edge magnitude $H_{0,i}$ is the measured BACI response for three monitored cuts (CEH36 +130 mm, CEH21 +74 mm, CEH18 +9 mm; Section 4.5) and assumed equal to the CEH36 value for the five unmonitored cuts. Scrape interiors are masked — each cut slack rises (Section 5.4.1) — so the field shown is the surrounding drawdown, the dipole counterpart of the slack rise. The modelled drawdown reach is not independently resolved in the 88-well network (Script 09b, $p=0.54$). Steady-state construction; equilibration over years to decades (Section 5.4.3). Clipped to the study-area outline; well markers coloured by cluster. Source: (Source: 20_scrape_drawdown_nohead.png).

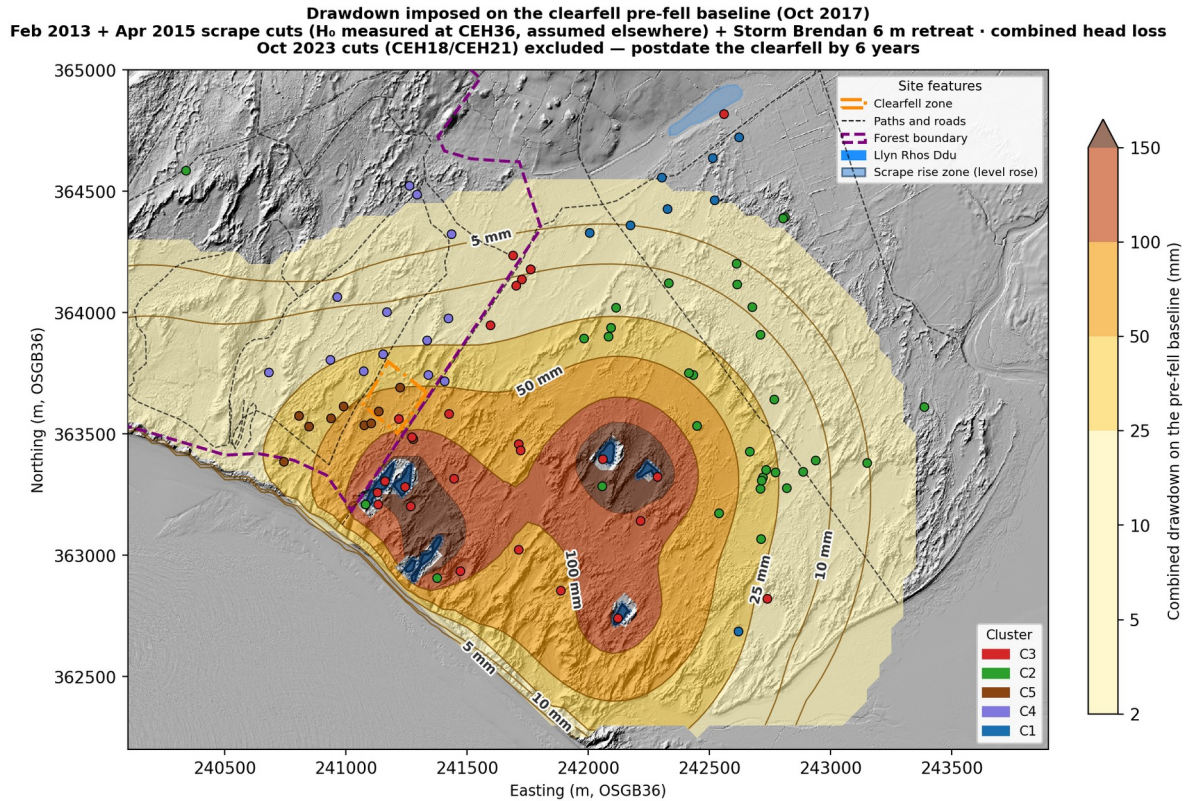


Figure 1.1.16: Combined drawdown imposed on the clearfell pre-fell baseline (Oct 2017): the cell-by-cell sum of the Feb 2013 and Apr 2015 dune-scrape cuts (five cuts; H_0 measured at CEH36 +130 mm, assumed = CEH36 for the four unmonitored cuts; Oct 2023 cuts excluded — they postdate the clearfell by six years) and a single Storm-Brendan-scale coastal-retreat event (Figure 52), both head losses, on the shared sequential drawdown scale. The scrape component uses leaky-aquifer superposition with method-of-images coastal boundary correction (see Figure 55); the coastal-retreat component is illustrative at the Storm Brendan scale. Sea-level rise is excluded; the field is the spatial footprint of the two non-felling drivers the clearfell BACI must separate from the felling signal (Sections 4.6, 5.5), not a coastal mass balance. A first-order superposition, not a closed water-balance solution. Clipped to the study-area outline; well markers coloured by cluster. (Source: 20_clearfell_baseline_drawdown.png).

1.2.5 Per-Well Forest Zone Analysis

The spatial coefficient atlas (Figure 48b–d) shows that the C4 and C5 forest wells span nearly the full site-wide range of β_2 values despite sharing the same Corsican pine canopy. To test whether this within-forest variation tracks topographic position (a proxy for the uncored substrate) or canopy differences, per-well SSM coefficients for the 14 forest zone wells (9 in C4, 5 in C5) were regressed against elevation, distance from the ridge and Easting (Section 3.4.4).

Elevation is overwhelmingly the dominant spatial predictor (Table 20). The atmospheric draw coefficient β_2 shows the strongest relationship ($r = 0.981$, $p < 0.001$; $R^2 = 0.963$), with higher-elevation wells on the ridge returning systematically higher β_2 values. Adding distance from the ridge contributes virtually no additional explanatory power ($R^2 = 0.963$). The drainage coefficient β_3 shows a significant but weaker elevation dependence ($r = -0.787$, $p < 0.001$; $R^2 = 0.619$), with distance from the ridge adding meaningful information (combined $R^2 = 0.739$): wells further from the ridge and at lower elevation drain faster. The recharge coefficient β_1 shows no significant relationship with elevation ($r = 0.258$, $p = 0.374$); Easting is the best single predictor ($r = 0.620$, $p = 0.018$) but explains only 38% of variance. This elevation–drainage relationship is visible directly in the drainage decay half-life map (Figure 49), where C4 wells reach $t_{1/2}$ values of up to 88 months compared with 12–19 months at C5.

C4 and C5 form two clearly distinct groups rather than a continuum. They separate almost entirely on the β_2 axis: C4 wells cluster at $\beta_2 = 2.2$ – 3.7 while C5 wells sit at $\beta_2 = 0.9$ – 1.5 , with zero overlap. Mean elevations differ significantly (C4: 10.7 m, C5: 5.7 m; $p < 0.001$), with a clean gap between 6.5 m and 8.7 m containing no wells. This corresponds to the transition from the elevated ridge and plateau terrain — where a deep unsaturated zone allows strong atmospheric draw and weak drainage — to the low-lying coastal fringe, where the shallow water table limits atmospheric influence and enhances lateral drainage. Both zones carry Corsican pine canopy, so the separation tracks elevation and water-table depth rather than canopy: C4 sits on the flank of the bedrock ridge (which outcrops within the forest), C5 on the lower coastal sand, with the depth and extent of rock beneath each unconstrained without coring.

Two individual wells merit comment. CEH14, at the highest elevation in the forest zone (14.4 m), has the highest β_2 in the network (3.723) — entirely consistent with the elevation regression rather than being an outlier. Its negative β_3 (-0.016) and low R^2 (0.568) are not well identified at this ridge-proximal position. The estimate is not significantly different from zero ($p = 0.24$), and six of the nine C4 wells likewise carry a non-significant β_3 , so CEH14 is at the imprecise end of a cluster that is imprecise throughout rather than a categorically different well. It overlies irregular bedrock topography on the elevated ridge flank, where buried ridges impede lateral drainage and head-dependent changes in saturated area would weaken or invert the apparent drainage response. NW10, the broadleaf block

well, has the highest β_1 in the forest zone (3.484, 2.4 SD above the C4 mean) and the highest β_3 among C4 wells (0.043). Its elevated recharge sensitivity likely reflects the deciduous canopy intercepting less winter rainfall than the surrounding Corsican pine (15% vs 24% interception), though this cannot be confirmed with a single well.

Coefficient	vs Elevation r (p)	vs Dist. ridge r (p)	vs Easting r (p)
β_1 (recharge)	0.258 (0.374)	−0.561 (0.037)	0.620 (0.018)
β_2 (atm. draw)	0.981 (<0.001)	−0.900 (<0.001)	0.749 (0.002)
β_3 (drainage)	−0.787 (0.001)	0.596 (0.024)	−0.419 (0.136)

Table 1.1.5: Per-well spatial predictors of SSM coefficient variation within the forest zone (C4 Main Forest and C5 Coastal Forest, $n = 14$). Pearson correlation coefficients and p -values for three spatial variables. Elevation dominates β_2 ($R^2 = 0.963$); β_1 shows no significant elevation dependence. (Source: 10c_04_forest_zone_summary.txt).

The per-well analysis confirms that the C4/C5 partition in the Ward's solution tracks a genuine elevation discontinuity rather than an arbitrary split within a continuous canopy population. On the topographic and coefficient evidence this is best read as a substrate-and-elevation boundary; the bedrock the forest overlies outcrops within the plantation, but its depth and extent across the C4/C5 break are not constrained without coring. The mechanistic interpretation of this finding — and its implications for forest management — is developed in Section 5.6.1.

1.2.6 Mean Water Table Surface and Groundwater Flow Direction

The mean annual water table surface was constructed by IDW interpolation of per-well mean maOD values (2005–2026) to a 50 m grid on the British National Grid, masked to the rectangular sea-boundary extent (Figure 57; Section 3.8.4). No DEM-height mask is applied because the water table is a continuous surface independent of the overlying topography. The highest mean water levels (~14 m AOD) occur in the north-western forest zone adjacent to the bedrock ridge, declining to 7–9 m AOD across the central dune and reaching 2–3 m AOD at the southern and eastern coastal margins. Contours run approximately NE–SW across the site, consistent with the topographic configuration. normalized Darcy flow direction vectors, computed as $Q = -K \cdot b \cdot \nabla h$ (Bear, 1972; Freeze and Cherry, 1979) from the head gradient field using $K = 6.0 \text{ m day}^{-1}$ (Betson et al., 2002), show a radial pattern diverging from the north-western high point: predominantly southward and south-westward in the western half of the site, and south-eastward in the eastern half. This pattern is broadly consistent with the DEM-derived flow network (blue lines in Figure 57; Conrad et al., 2015; Natural Resources Wales LiDAR DTM, 2023). The flow vectors are reported as directional context for the interpolated head field rather than as output of a calibrated continuous-flow model.

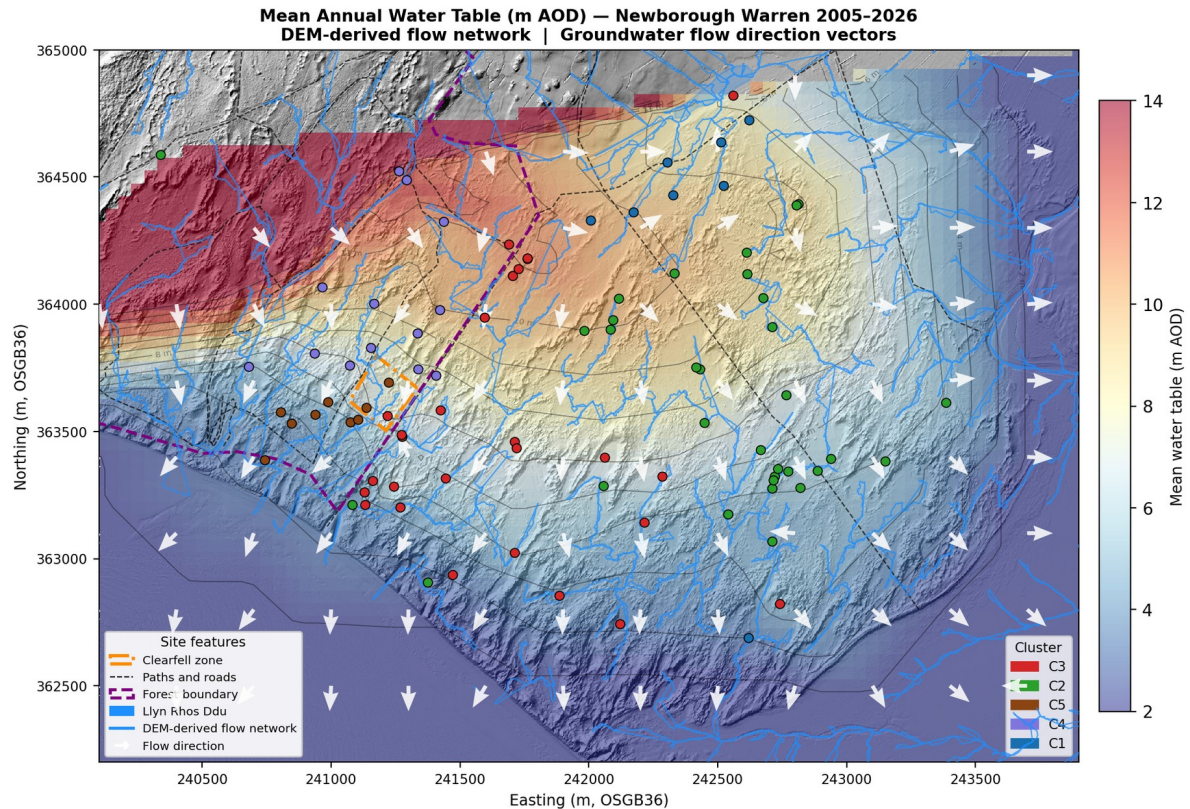


Figure 1.1.17: Mean annual water table elevation (m AOD) across the Newborough Warren dune aquifer, 2005–2026, derived by IDW interpolation of per-well mean annual heads to a 50 m grid. The highest water levels (~14 m AOD) occur in the north-western forest zone adjacent to the bedrock ridge, declining toward the southern and eastern coastal margins (~2–3 m AOD). Normalized groundwater flow direction vectors (white arrows; direction only, magnitude not scaled) are derived from the head gradient field $Q = -K \cdot b \cdot \nabla h$, using $K = 6.0$ m/day (Betson et al., 2002) and the IDW aquifer thickness surface. Well symbols are coloured by cluster. Background: LiDAR greyscale hillshade (LiDAR data: © NRW & OS; see Section 10 for full attribution). Generated by `20_spatial_figures.py`. (Source: `20_head_surface_streams.png`)

1.2.7 Water Balance Residual Field

The SSM water balance residual field (Figure 58; Section 3.8.4) quantifies the degree to which the three-term SSM closes the long-term water balance at each well. The residual at each reference well is computed as $\alpha = (\beta_2 \cdot \overline{PE\bar{T}} + \beta_3 \cdot \bar{h}_{\text{disp}}) - \beta_1 \cdot \bar{P}$, where overbars denote long-term monthly means, $\bar{h}_{\text{disp}}$ is mean displacement above the 3.7 m drainage datum, and $\bar{P}$ is gross rainfall. The averaging window is taken from the mean-head record itself (March 2005 to February 2026, $n = 252$ months; $\bar{P} = 74.33$ mm/month, $\overline{PE\bar{T}} = 54.72$ mm/month), so that the climate means and the mean heads describe the same period.

Gross rather than interception-corrected rainfall is used at forest wells. Script 03 fits the SSM on gross rainfall and above-canopy Thornthwaite PET, so the canopy interception loss is already carried inside the fitted $\beta_2 \cdot \overline{PE\bar{T}}$ term; reducing $\bar{P}$ as well would subtract it twice. Interception is a partition of the atmospheric energy budget rather than a term additive to it (Section 3.8).

Positive values indicate locations where modelled losses exceed modelled recharge, implying an unmodelled input; negative values indicate the converse. The field closes tightly and carries no coherent spatial structure, with a small systematic negative offset. Residuals span -0.0115 to $+0.0049$ m/month about a median of -0.0045 m/month (mean -0.0038), with 64 of the 66 reference wells falling within ± 0.01 m/month and 58 negative. All five cluster medians are negative, ranging from -0.0065 m/month at C4 Main Forest to -0.0004 m/month at C5 Coastal Forest, all small relative to the coefficient uncertainties that generate them. No DEM-height mask is applied. Residual magnitude is uncorrelated with position on either axis (Spearman $\rho = +0.099$ against Easting, $p = 0.43$; $\rho = +0.111$ against Northing, $p = 0.37$). The signed residual carries a weak west-to-east gradient that falls short of conventional significance ($\rho = -0.226$ against Easting, $p = 0.07$; $\rho = -0.171$ against Northing, $p = 0.17$); it is reported for completeness rather than as an established spatial structure.

No well exceeds $+0.02$ m/month, and only eight of the sixty-six carry a positive residual at all. The largest positive values are CEH4 ($+0.0049$), NW7 ($+0.0033$) and NW6 ($+0.0032$) in C3 Western Residual, with NW9 ($+0.0032$) and CEH17 ($+0.0031$) in C5 Coastal Forest — all in the open dune or at the coastal forest margin rather than at the ridge. No C4 Main Forest well is positive. The largest negative residual occurs at the open-dune well D7 (-0.0115 m/month), with the ridge-flank well CEH14 (-0.0106 m/month) close behind; at both, modelled recharge marginally exceeds modelled losses.

The residual field is therefore a diagnostic of model adequacy rather than a map of unmeasured lateral flux. Across all five clusters the three-term SSM closes the long-term water balance without requiring an additional input term, and the corrected field agrees with the independent water balance of Script 16, which computes the same closure on gross rainfall over the study period and agrees to within ± 5.8 mm/month at every cluster, the largest discrepancy falling at C4 Main Forest. The independently computed Darcy flow vectors derived from the mean head gradient field indicate a radial flux diverging from the north-western high point — southward and south-westward in the western half of the site, south-eastward in the eastern half (Section 4.9.6); they are retained as directional context for the site's flow geometry and are not evidence bearing on the residual field, which shows no ridge-proximal signal.

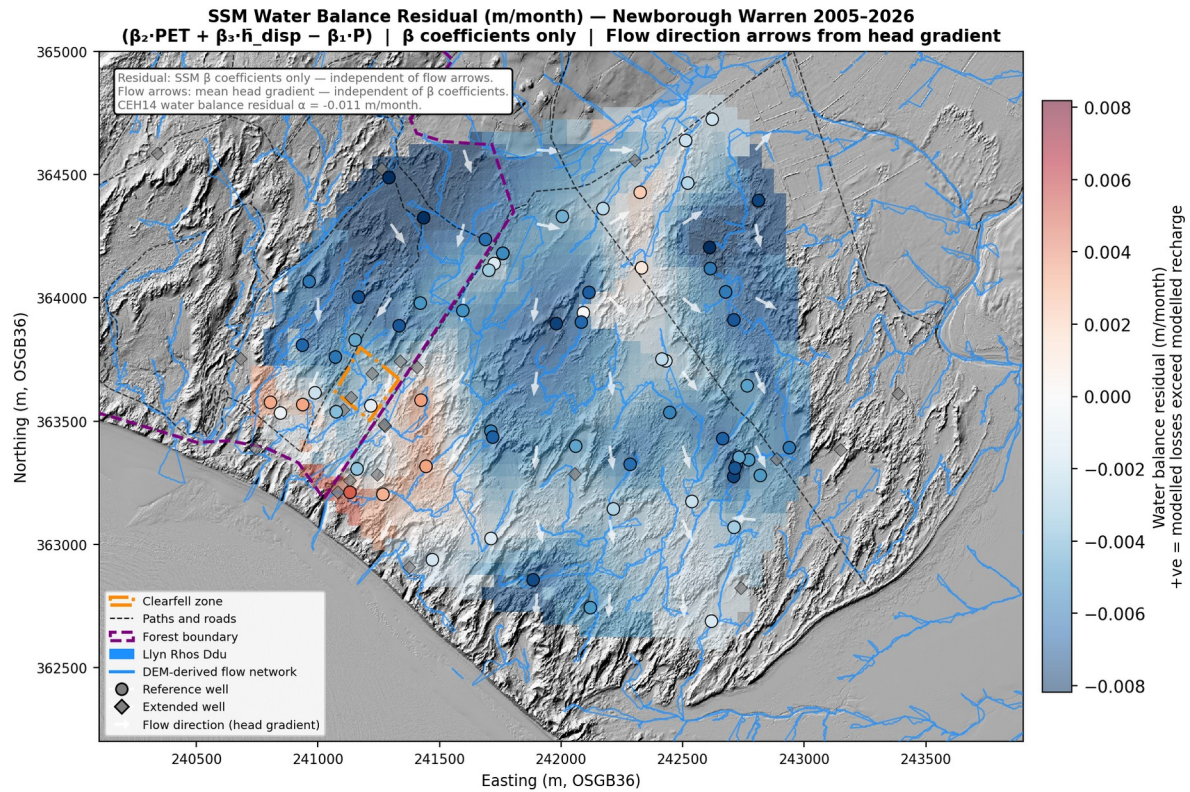


Figure 1.1.18: SSM water balance residual across the Newborough Warren dune aquifer, 2005–2026. The residual at each reference well is computed from β coefficients only as $(\beta_2 \cdot PET + \beta_3 \cdot h_{disp}) - \beta_1 \cdot P$, using gross rainfall, and IDW-interpolated to a 50 m grid, clipped to the site boundary and to 100 m beyond the outermost dipwell. Climate means are taken over the same period as the mean heads (March 2005 to February 2026, $n = 252$ months). Positive values (red) indicate locations where modelled losses exceed modelled recharge; negative values (blue) the converse. The field closes tightly and shows no coherent spatial structure, with a small systematic negative offset: residuals span -0.0115 to $+0.0049$ m/month about a median of -0.0045 m/month, with 64 of the 66 reference wells within ± 0.01 m/month and 58 negative. No well exceeds $+0.02$ m/month; the largest positive value is CEH4 ($+0.0049$ m/month) in C3 Western Residual. The largest negative residual is at the open-dune well D7 (-0.0115 m/month), with CEH14 (-0.0106 m/month) close behind. White arrows show normalized Darcy flow direction vectors derived from the mean head gradient field (independent of the β coefficient residual calculation). Well symbols are coloured by residual value; circles indicate reference wells, diamonds indicate extended wells. Background: LiDAR greyscale hillshade. (LiDAR data: © NRW & OS; see Section 10 for full attribution.) Generated by `20_spatial_figures.py` (Source: `20_residual_ssm.png`).

1.2.8 Combined Driver Assessment and Observed Network Change

Four further spatial figures are derived from the framework developed in Sections 4.9.3 and 4.9.4: a measured clearfell step-change map (Figure 59), a parametric net state map superimposing all five active drivers (Figure 60), and an empirical MSL5 change map comparing the pre-clearfell and current MSL5 windows (Figure 62) and a modelled 2005–2025 driver-change map (Figure 61). Methods for all three are in Section 3.8.5; interpretation of the comparison between Figures 59 and 60 is in Section 5.7.5.

Measured clearfell spatial response

The clearfell step-change map (Figure 59) shows the difference in mean water-table level between the pre-felling scraping era and the post-felling era at each well (*fell_step_cc*, Script 10b; positive = shallower after felling). The map shows a broadly positive response across the open dune interior (+25 to +100 mm), with the largest gains at the C4 Forest Control wells immediately adjacent to the felled compartment. A zone of -25 to -100 mm appears along the south-western coastal margin at the C5 Coastal Forest and southern C3 wells. This pattern is consistent with the coastal-retreat gradient established in Section 4.8.2 and mirrors the Coastal Control tier deterioration of Section 4.6.4; it does not reflect a clearfell response. The spatial distribution corroborates the BACI finding of Section 4.6.3 that the felling signal is concentrated in the C4 and immediately adjacent C3 interior, not at the coastal margin.

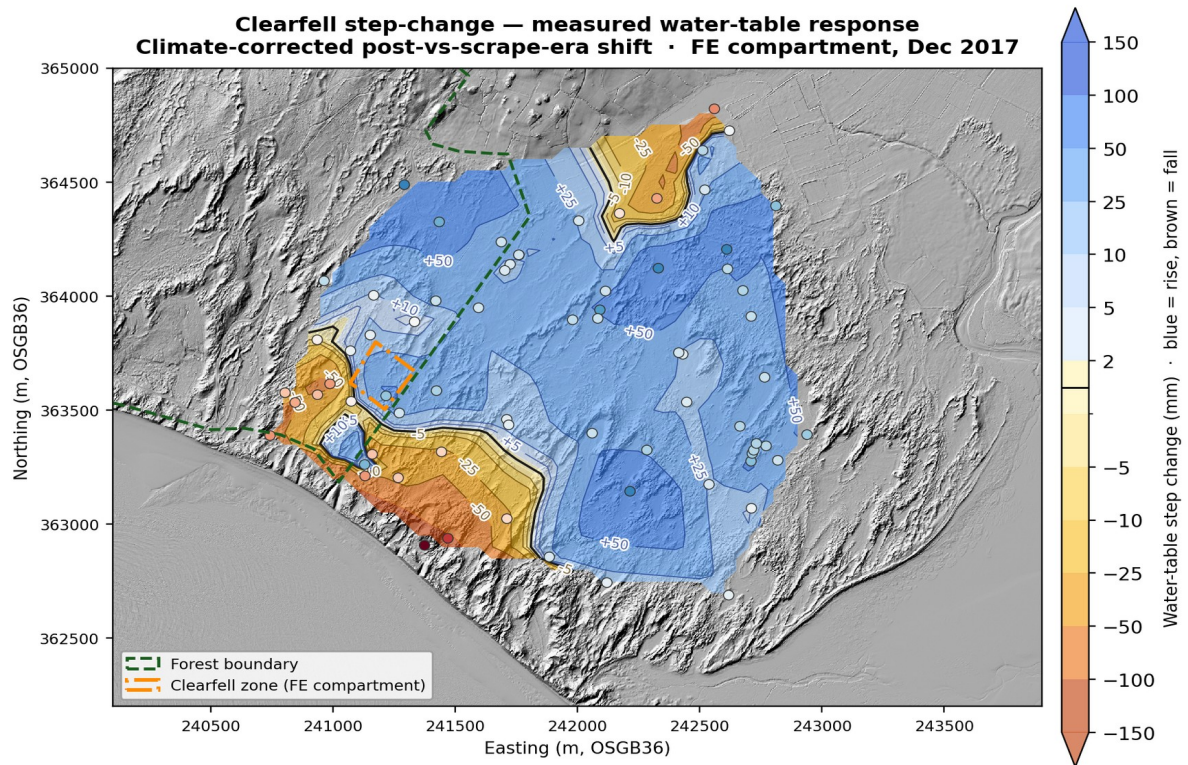


Figure 1.1.19: Clearfell step-change map — climate-corrected mean water-table shift, pre-felling (scraping era) versus post-felling, at each well. Derived from column `fell_step_cc` (Script 10b; Section 3.5.4). Positive values (blue) = shallower after felling. Wells excluded where the post-felling record is too short to estimate a stable era mean (<6 post-felling monthly observations); $n = 70$ wells retained. Both interpolated surface (IDW power 1.5, Gaussian $\sigma = 100$ m) and well markers use the same discrete diverging colourmap. Forest boundary (green dashed), clearfell zone (orange dash-dot). Generated by `20_spatial_figures.py` (Source: `20_clearfell_gain.png`).

Parametric net state map

Figure 60 (method: Section 3.8.5) superimposes all five annual-mean driver fields on the site grid. The forest interior carries a large net loss (−50 to over −150 mm) where the standing canopy drawdown dominates. A small net-gain zone sits over the felled compartment where the clearfell interception recovery locally exceeds the surrounding forest loss. The open dune is near-neutral: neither the erosion drawdown nor the SLR head gain reaches the eastern dune interior at meaningful magnitude. Along the southern shoreline, a narrow net-gain band reflects SLR marginally exceeding the Storm Brendan erosion drawdown at the immediate dune toe.

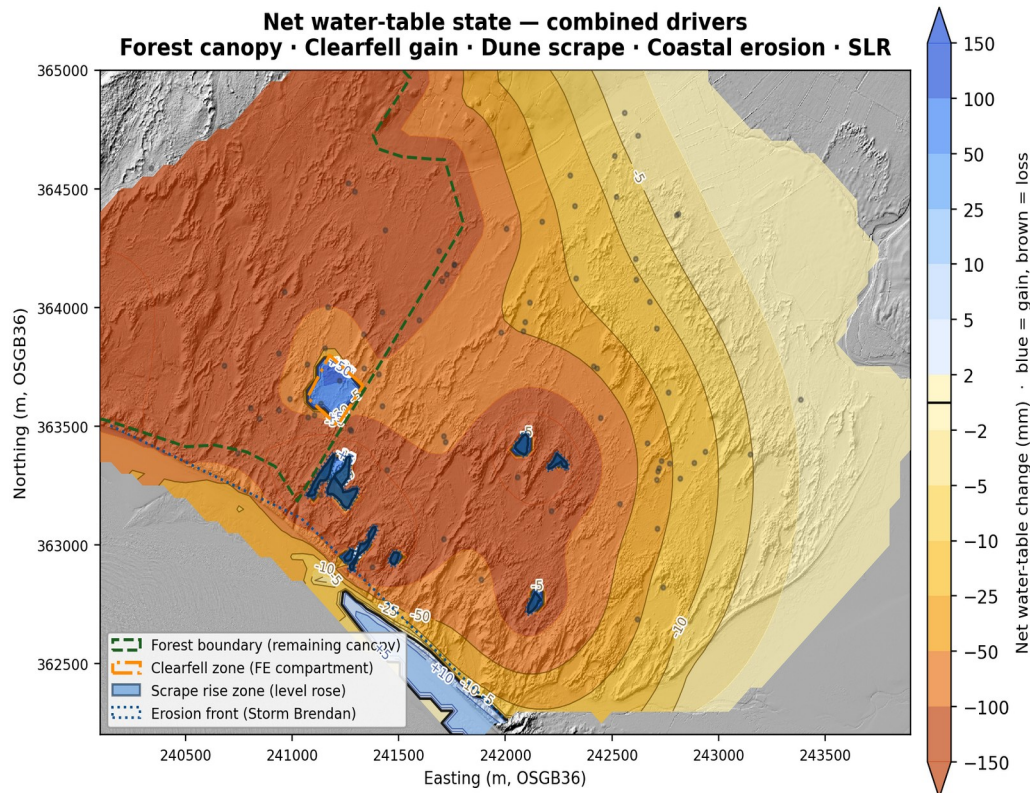


Figure 1.1.20: Net water-table state — parametric combination of five simultaneous annual-mean drivers: forest canopy drawdown ($H_0 = 150$ mm, $\lambda \approx 230$ m), clearfell gain (same H_0 , λ), dune-scrape contribution (all eight cuts; H_0 measured at CEH36 +130 mm, CEH21 +74 mm, CEH18 +9 mm; others assumed = CEH36; leaky-aquifer superposition with method-of-images coastal correction; scrape footprints contribute + H_0 rise, surrounding area contributes drawdown), coastal erosion ($h_0 = 21$ mm), SLR (+20 mm over 5 yr). Net = SLR gain + clearfell gain + scrape rise − forest loss − erosion loss − scrape drawdown. Symmetric ± 150 mm diverging scale; bold zero contour. First-order superposition of annual-mean fields; not a closed water balance (caveats: Section 4.9.4). (Source: 20_net_state_map.png).

The 2005 → 2025 driver-change map

The 2005 → 2025 driver-change map (Figure 61) re-bases the parametric net-state construction to the study record, differencing each modelled driver field between its 2005 and 2025 states so that the map shows what the drivers imply should have changed over the monitoring window rather than the accumulated state since the forest was planted. Because the standing pine canopy is at equilibrium at both endpoints, its drawdown cancels in the difference and the mature forest interior returns to near zero — the check that the map is an epoch difference and not a re-drawing of Figure 60. Five fields change over the window and are summed: the clearfell gain, computed as in the net-state map from the removal of the interception deficit over the felled compartment; the dune-scape response, with all cuts within the window contributing; the five-year chronic coastal drawdown; sea-level-rise head gain over the same span; and a broadleaf-restock field new to this map. The broadleaf-restock block, felled in 1993 and restocked in 1998, was already near closed canopy at the 2005 baseline and reaches full canopy by 2025; only the increment of interception deficit that develops over 2005 → 2025 enters the map, computed on the same exponential-decay construction as the forest field but scaled by the broadleaf interception fraction (0.15, against 0.24 for pine; Komatsu et al., 2011) and by the canopy-establishment increment, giving of order 56 mm of added drawdown at source. It appears as a drawdown patch of the opposite sign to the clearfell gain — the atmospheric-draw signature of a canopy returning where the clearfell removed one. The coastal-retreat field is expressed as a five-year chronic drawdown, five years of the observed coast-edge decline rate (from the live Script 25 forest-free fit, $\delta_0 = 29.0 \text{ mm yr}^{-1}$), giving $h_0 = 5 \times \delta_0 \approx 145 \text{ mm}$ at the dune toe on the same five-year horizon as the sea-level-rise field — the strongest of the five fields at the western coastal margin, and rate-independent by construction. As a first-order superposition of single-mechanism fields the map is illustrative and not a closed water balance; the broadleaf and coastal fields in particular are modelled quantities the dipwell record cannot fully resolve, and are presented as indicative. The map is the modelled companion to the observed climate-removed trend map (Figure 64, Section 3.8) over the same 2005 → 2025 window: the two are built from different data by different methods and are expected to agree in broad direction rather than in detail, so their agreement where it occurs supports the attribution and their divergence flags a driver the model does not capture.

Newborough Warren — MODELLED driver change, 2005 → 2025 · log colour scale
Clearfell gain · Broadleaf restock · Dune scrapes · 20-yr chronic coastal drawdown · SLR

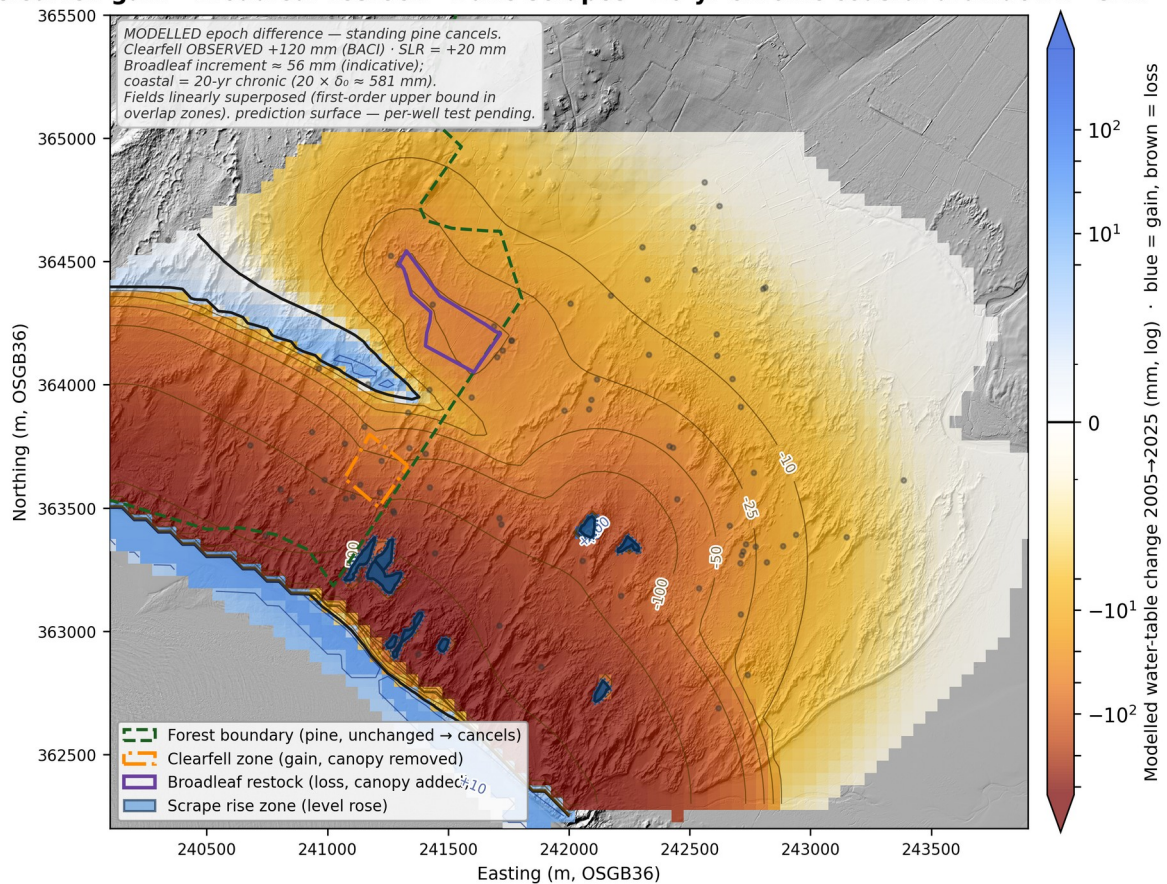


Figure 1.1.21: Modelled water-table change across Newborough Warren, 2005–2025 — full-window (20-year) coastal horizon, logarithmic scale. The companion to the five-year driver-change map, showing the same modelled drivers but with the coastal field accumulated over the full study window. The map is the epoch difference of the modelled driver fields (2025 minus 2005): the standing pine canopy is at equilibrium in both epochs and cancels, so the near-white forest interior is the check that the difference is working rather than a re-drawing of the standing-forest drawdown. The drivers that change over the window are the December 2017 clearfell (interception deficit removed → head gain, blue, plotted at the observed BACI step of +120 mm), the broadleaf restock (canopy interception added → head loss, an indicative +56 mm increment as the block approached closed canopy), the 2015 and 2023 dune scrapes (slack rises with surrounding drawdown cones), sea-level rise (+20 mm), and the coastal chronic decline. The coastal field here is the full-window accumulation of the fitted coast-edge trend ($20 \times \delta_0 \approx 581$ mm at the dune toe, decaying to zero at the reach $L = 894$ m); the five-year companion instead shows the near-term chronic value (≈ 145 mm) for a legible comparison of the management drivers. Because the coastal drawdown (of order several hundred millimetres) is far larger than the management effects (of order one hundred millimetres), the colour scale is logarithmic so both read on one map: deep red marks the strongest coastal drying at the toe, fading through amber to white where change is negligible, and blue marks head gain. The fields are combined by linear superposition; where they overlap the map shows their sum, which is a first-order upper bound and may over-state the true combined drawdown. The map is a modelled prediction surface — at each well it predicts a 2005–2025 change that can be tested against the observed record (a per-well validation is reported separately); the coastal and broadleaf fields are the least record-constrained and are flagged as modelled. Inland propagation of the coastal signal toward the reach is diffusive and develops over years to decades; the magnitudes shown are the accumulated end-of-window values, not the present-day resolved field. (Source: 20_driver_change_2005_2025_20yr.png)

Observed network change

Figure 62 presents the conventional two-window reading of observed network change — validated on its own terms (Section 5.7.5) but, as set out below, limited in what a two-window difference can establish about absolute change.

The MSL5 change map (Figure 62; method: Section 3.8.5) compares window end 2017 (springs 2013–2017, pre-clearfell, network mean –492 mm below ground) with window end 2023 (springs 2019–2023, network mean –589 mm). The raw 2023-minus-2017 difference exceeds the ± 25 mm threshold at 56 of 59 wells, every one of them deepening; no well shallows. The map shows a broadly deeper aquifer in the current window, the deepening intensifying from the eastern dune interior toward the south-western coastal margin. Because the figure differences two five-year windows, every time-invariant driver cancels — in particular the constant drawdown of the 60-year standing forest, which is present in both windows and contributes nothing to the difference — so the surviving pattern reflects only drivers that progressed across the study period, principally the coastal influence developed in Section 4.8.2; it is not, and cannot be, the standing-canopy drawdown shown parametrically in Figure 60.

The eastern C1/C2 dune wells deepen least (CEH25 –13, L7 –17, D7 –25 mm), several falling within the ± 25 mm near-neutral band. The clearfell Impact well WMC3 (–85 mm) and the scrape well CEH36 (–74 mm) deepen broadly in line with the –97 mm site-wide mean, showing no intervention-specific departure from the surrounding network. The most strongly negative zone is the south-western coastal margin: CEH22 (–229 mm), CEH21 (–194 mm), CEH18 (–158 mm), and the C5 coastal members CEH19, CEH17, NW9 (–150 to –153 mm). These wells are most directly exposed to the coastal-retreat gradient documented in Section 4.8.2, and their deepening is consistent with the BACI easting $\times$ time absorption of Section 4.6.3.

The C4 Forest Control wells deepen by –130 mm (cluster mean), more than the –97 mm site-wide average. As noted, the constant drawdown of the 60-year standing forest cancels in a window-to-window difference, and these are unfelled controls, so this excess is neither a standing-canopy nor a clearfell effect; it is consistent with the forested clusters' stronger differential response to climate, whereby their higher β_2 atmospheric draw and 24% canopy interception convert the between-window climate anomaly into a larger water-level change than the open dune. Because the map applies no climate correction, this differential response is expressed in full: the forest-shaped deepening is therefore a climate-response artefact of the uncorrected differencing, not an active forest signal.

The interannual standard deviation of annual spring level differs widely across the network: about 175 mm at C1, 252 mm at C2, 315 mm at C3, 277 mm at C5 and 404 mm at C4, giving a five-year window mean a standard error of about 78, 112, 141, 124 and 181 mm respectively. Controlling for the drainage recession time, this standard deviation correlates

with the atmospheric-draw coefficient β_2 at partial rank $r = +0.63$ ($p < 0.0001$); controlling for β_2 , its association with the recession time is not significant (partial $r = +0.16$, $p = 0.20$). The two parameters are themselves correlated across wells (Spearman $+0.37$). Across the 65 wells with at least twelve valid spring years, the lag-1 autocorrelation of annual spring level is approximately zero at every cluster, with cluster means between -0.14 and $+0.05$, against an expectation of 0.30 at C1 rising to 0.77 at C4 if the fitted drainage term is read as a first-order annual recession; it is uncorrelated with the recession time (Spearman $+0.05$, $p = 0.70$), and the site-mean spring series over 21 years gives -0.00 .

These results are robust within the two-window method: the spring-gap and independent-metric checks of Section 5.7.5 indicate the 2017 → 2023 deepening is not an artefact of window length or spring-record gaps. Their limitation is comparative rather than internal. Because the amplitude of interannual variation differs so widely across the network, the absolute magnitude of a two-window difference — and at some wells its sign — depends on which two windows are compared, even between windows matched for antecedent rainfall; the matched-window sensitivity analysis of Section 5.7.5 yields site-mean changes spanning -0.14 to $+0.22$ m and changing sign across all admissible window pairs (19 of 66 falling, 47 rising) — the -97 mm comparison being one interior point and the wet-2024 windows the $+0.22$ m extreme — which is why absolute site-wide change is not rested on this two-window method (Script 34). The -97 mm site-wide figure is therefore specific to the 2017/2023 comparison, not a robust estimate of secular change. The two figures that follow remove this dependence: Figure 63 maps the differential movement with the common climate signal removed, and Figure 65 the climate-response envelope — the latter quantifying directly the forested clusters' larger climate response invoked above.

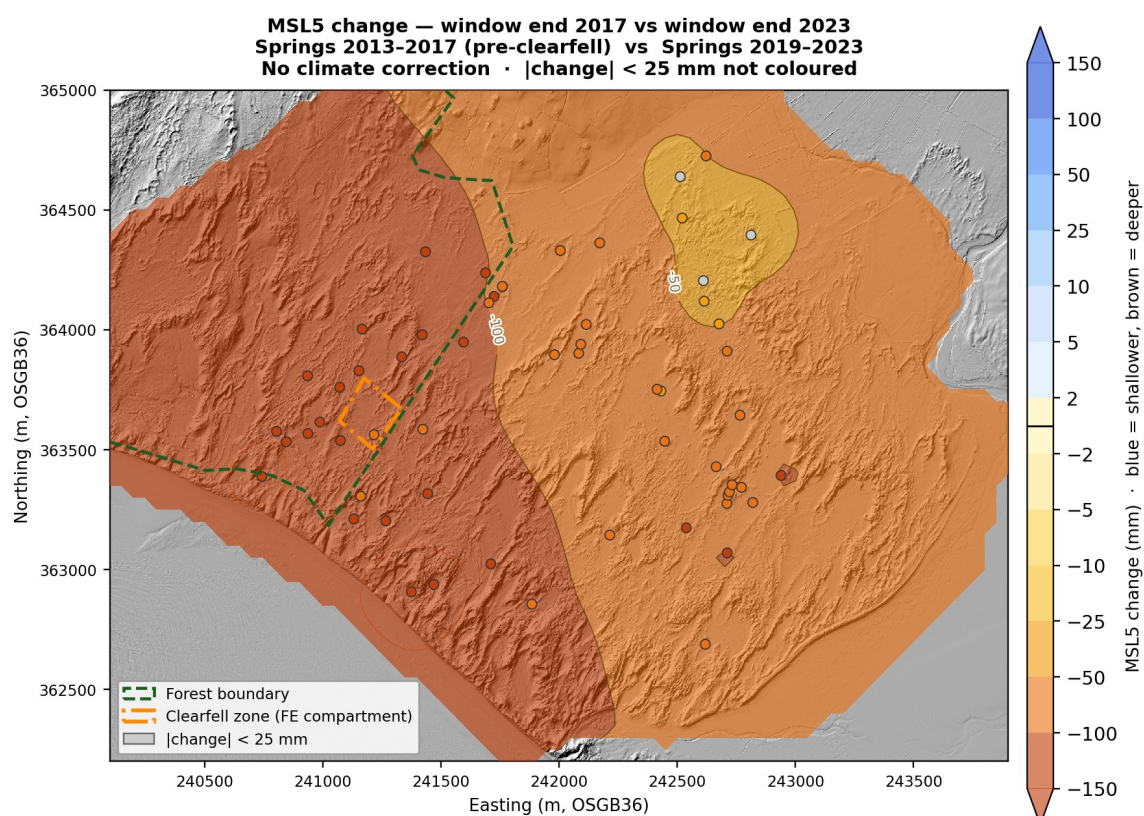


Figure 1.1.22: Observed MSL5 change: window end 2017 (springs 2013–2017, pre-clearfell) vs window end 2023 (springs 2019–2023). Raw difference in mm (positive = shallower); no climate correction or common-mode normalization. Network mean -492 mm (2017) to -589 mm (2023), below ground. Changes $< \pm 25$ mm not coloured; 56 of 59 wells exceed the threshold and all deepen. $n = 59$ wells. IDW power 2.0, Gaussian $\sigma = 50$ m; well markers use same discrete banded colourmap as surface. Window end 2025 (springs 2021–2025) was examined but not adopted: it contains the two wettest springs of the record, and the resulting Forest Control excess is not separable from window composition (Section 5.7.5). The two C4 wells CEH13 and CEH14 are excluded separately, on a near-zero and a negative SSM drainage coefficient respectively, which makes their five-year window values unreliable. As a window-to-window difference, the map cancels time-invariant drivers — including the standing-forest drawdown of Figure 50; the forested zone's appearance is a climate-response artefact, not a canopy signal (§4.9.8). (Source: 20_msl5_change_2017_2023.png).

Secular differential movement

Figure 63 shows the secular differential movement of the spring water table across Newborough Warren: with the common climate signal removed, the high, slow-draining forest mound holds its position while the fast-draining lake-edge and coastal margins decline relative to the site as a whole — a pattern governed by the aquifer's recession geometry rather than by management intervention. The quantity mapped is, for each well, the linear trend through its spring anomaly — the well's mean-spring level minus the site-mean spring level in the same year — over 2011–2025. Subtracting the site mean each year removes the shared climate forcing that dominates the raw record, so what remains is the rate at which each well is gaining or losing ground against the rest of the Warren. Because it is built from

a per-well trend rather than a two-window difference, the map is insensitive to the window-selection problem of Figure 62; it correlates above 0.98 with the two-window pattern but does not inherit its fragility.

The differential signal is spatially coherent but statistically uneven. Of the 72 wells mapped over 2011–2025, eight reach individual significance under an AR(1)-corrected trend test cross-checked by a moving-block bootstrap; these are predominantly the lake-edge and coastal wells that are losing ground. The forest-interior rise is coherent across neighbouring wells but individually sub-significant — each forest well is rising slowly against a noisy background — and it firms up over the longer 2005–2025 window (13 of 75 wells significant), consistent with a slow, persistent process rather than a recent step. Critically, the same holding-up behaviour appears in the C5 coastal-forest wells only where they sit on the high ground, not by virtue of their canopy: the inland C4 forest mound holds while C5 wells nearer the eroding coast decline with their open-dune neighbours. The pattern is therefore one of position within the recession geometry, not of forest cover, and it is not attributable to the felling intervention.

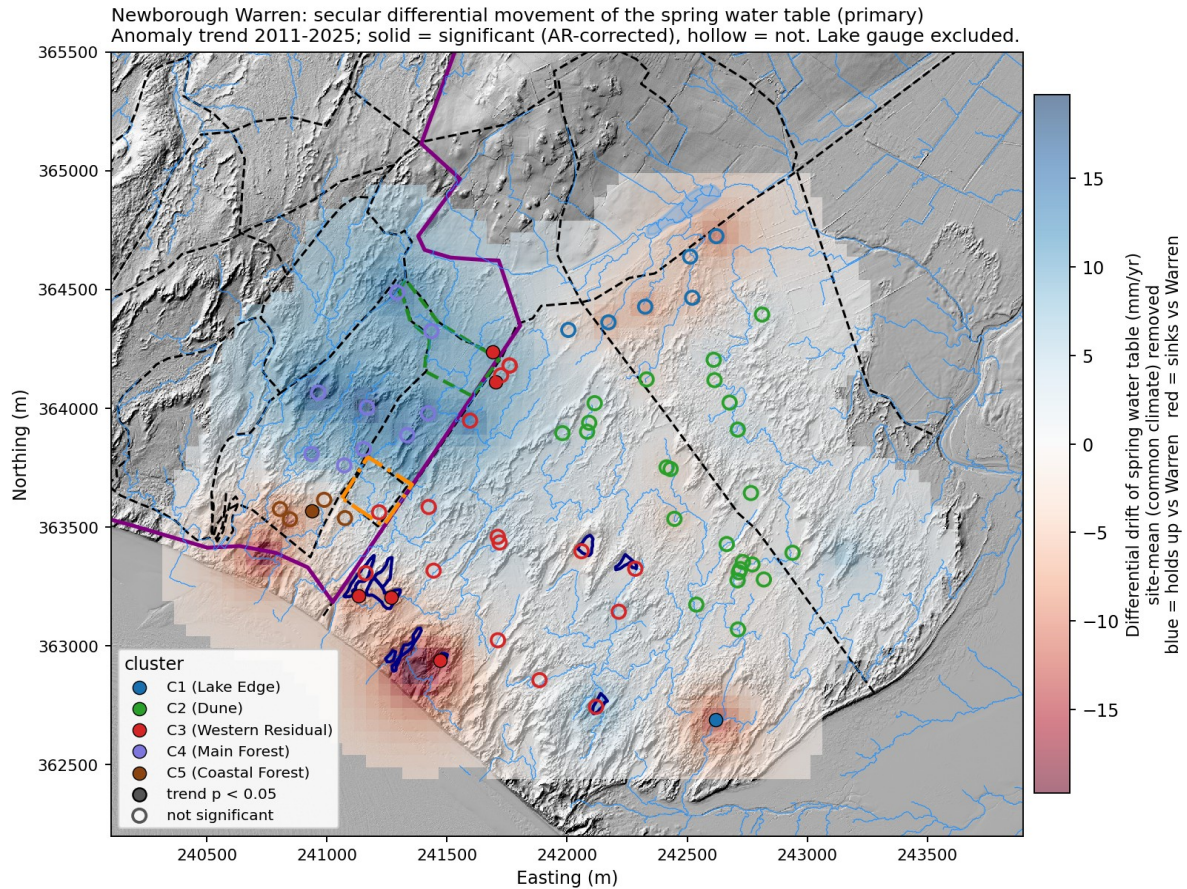


Figure 1.1.23: Secular differential movement of the spring (March, April, May) water table, 2011–2025. For each well the map shows the linear trend (mm yr^{-1}) of its spring anomaly, defined as the well's mean-spring level minus the site-mean spring level in the same year, so that the common climate signal is removed and only movement relative to the site remains. Read as a management diagnostic, it shows the direction and rate at which the spring water table is moving across the site over the study period - which areas are holding or gaining ground relative to the Warren as a whole, and which are losing it — the differential complement to the site-wide drying trend. Solid markers are significant under an AR(1)-corrected trend test corroborated by a moving-block bootstrap; hollow markers are not individually significant. Warm colours rise relative to the site, cool colours decline. CEH13 and CEH14 (SSM outliers) and the Llyn Rhos-Ddu lake gauge are excluded. (Source: 32_differential_movement_2011_2025.png)

Absolute movement with the climate signal removed

Figure 63 measures each well against the network mean, and over a wet-spring-lifted window that referencing can invert the plain reading: a well that is holding steady while the site mean rises appears to decline, and the slow-draining forest reads as *rising* not because its water table is recovering but because it is falling more slowly than the rest. Figure 64 removes that ambiguity by mapping the absolute secular trend at each well — the shared climate signal taken out by regression against an external climatic-water-balance index, but with no re-referencing to the network mean, so brown marks ground where the water table is genuinely falling and blue where it is genuinely holding or rising. The construction and window are described in Section 3.8.5; the trend is fitted over 2005–2025, the longer window being primary here because the shorter window is dominated by wells whose records begin only after 2011.

The absolute map tells a quieter and more spatially specific story than the differential one. Once the weather is removed, most of the Warren is close to holding its position — the lake-edge, dune and main-forest clusters all sit within a few millimetres per year of zero (C1 +2.1, C2 +4.6, C4 +2.7 mm yr⁻¹) — and the genuine, weather-independent losses are confined to the western and coastal margins. The coastal-forest cluster carries much the most negative trend (C5 $\approx$ -11.6 mm yr⁻¹), and the western residual cluster is mildly negative (C3 $\approx$ -3.4 mm yr⁻¹); the strongest individual driers are the western and coastal wells (CEH3 -28, NW9 -23, CEH4 -19 mm yr⁻¹). This is the same coastal-margin decline that Figure 63 identifies as the fastest-losing part of the network, but expressed in absolute terms: the forest interior is not recovering, it is simply stable, and the drying that matters is at the coast. The two maps are complementary — Figure 63 answers "which part of the Warren is gaining or losing ground against the rest," Figure 64 answers "where is the water table genuinely rising or falling" — and their agreement on the coastal margin, reached from different constructions, strengthens the reading that coastal retreat is the dominant secular driver.

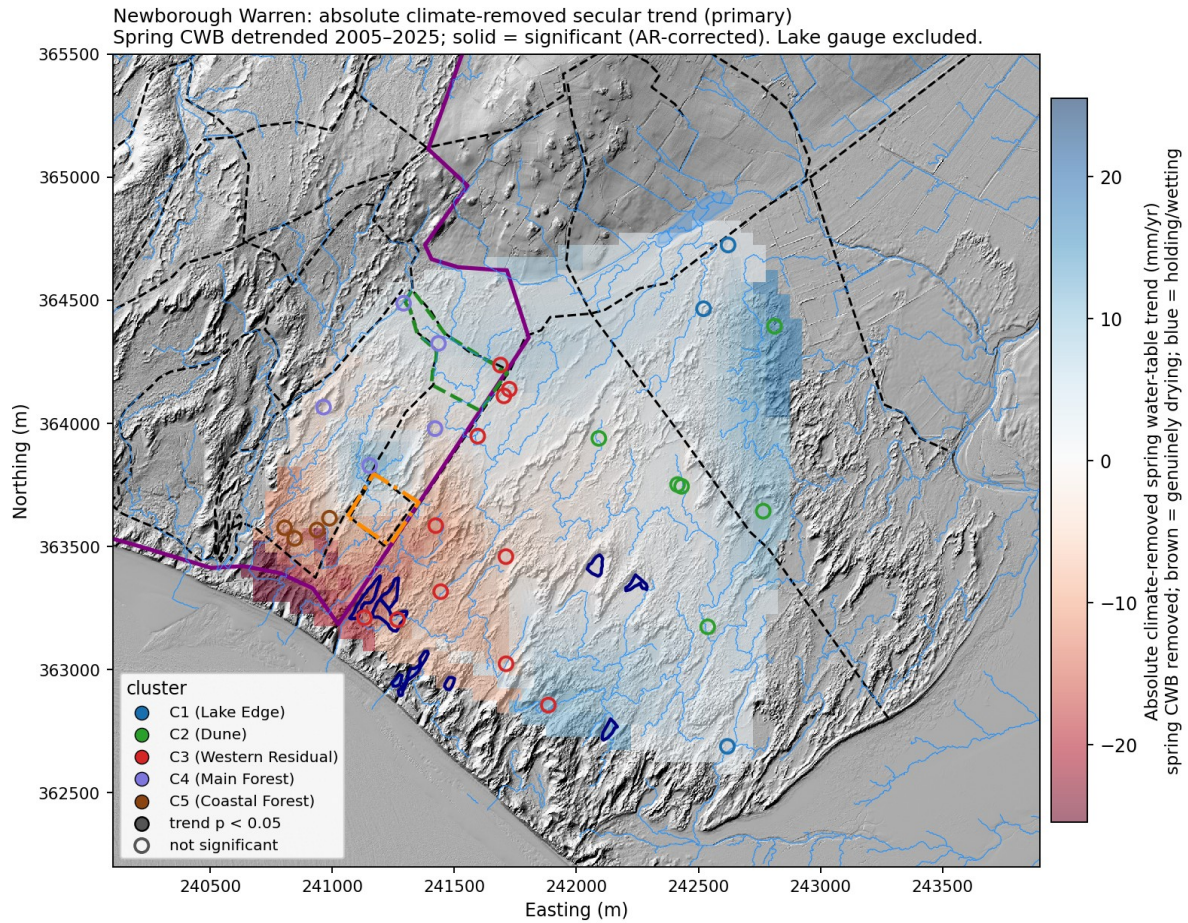


Figure 1.1.24: Absolute climate-removed secular trend of the spring (March, April, May) water table, 2005–2025. For each well the spring level is regressed jointly on an external spring climatic-water-balance index (RAF Valley) and a linear time term, and the fitted time coefficient is the secular trend in mm yr^{-1} , with the climate signal removed but without re-referencing to the network mean. Blue = genuinely rising or holding; brown = genuinely falling. Solid markers reach individual significance (AR(1)-corrected, bootstrap cross-checked); hollow do not. Wells lacking a genuine pre-window record are excluded (Section 3.8.5). The coastal and western margins carry the genuine secular losses; the interior, including the forest, holds once the weather is removed. (Source: 36_absolute_climate_trend_2005_2025.png.)

The climate-response envelope

Where Figure 63 measures slow secular change, Figure 65 measures climate sensitivity: how far each part of the network moves between the driest and wettest springs the record contains, and Figure 66 how close the dry-year water table already lies to ecological thresholds. The amplification panel (Figure 65) addresses the first. Between its antecedent-dry and antecedent-wet spring extremes the site-mean water table swings approximately 0.75 m; expressing each well's swing as a ratio to that network-mean swing, with the common-mode removed, gives a relative amplification field independent of window choice, because it compares true end-members rather than adjacent windows. The forest interior amplifies the shared swing to about 1.7×, while the lake edge damps it to about 0.6×, with the open dune close to unity; the forest is thus the most climate-volatile zone on the site and the lake margin the most stable. This is the same forest behaviour that inflates the C4 deepening in Figure 63, here measured directly rather than as an artefact of differencing. The amplification correlates most strongly with the atmospheric-draw coefficient β_2 ($r \approx +0.8$), which survives control for elevation and exclusion of the forest clusters; β_2 is best read as a marker of a deep, slow-recession store rather than a literal evaporation rate, since in C4 it is inflated by the known β_2 – β_3 degeneracy.

The second panel (Figure 66) maps the dry-year spring water-table depth — the mean spring level across the dry extreme years (2011, 2012, 2019). Spring is used in preference to the summer minimum because the spring record is near-complete whereas summer-minimum readings are sparse; the choice is deliberately conservative, since spring is the seasonal high and therefore the shallowest the water table reaches in those years. On this panel the Curreli et al. (2013) thresholds are overlaid not as summer-minimum crossings but to mark where even this spring high has already fallen below wet-slack (SD15b, 0.61 m) and dry-slack (SD16, 0.98 m) viability. Because the summer minimum lies deeper still, the contoured zones are a strict lower bound on the area below ecological viability: where the forest interior and western dune sit beyond SD16 even in spring, the water table is below dry-slack viability for the whole of those dry years. Together the two panels separate the two questions that the single two-window map of Figure 62 conflated — how sensitive each area is to climate, and how close it already sits to ecological thresholds in dry years — neither of which depends on an arbitrary choice of comparison window.

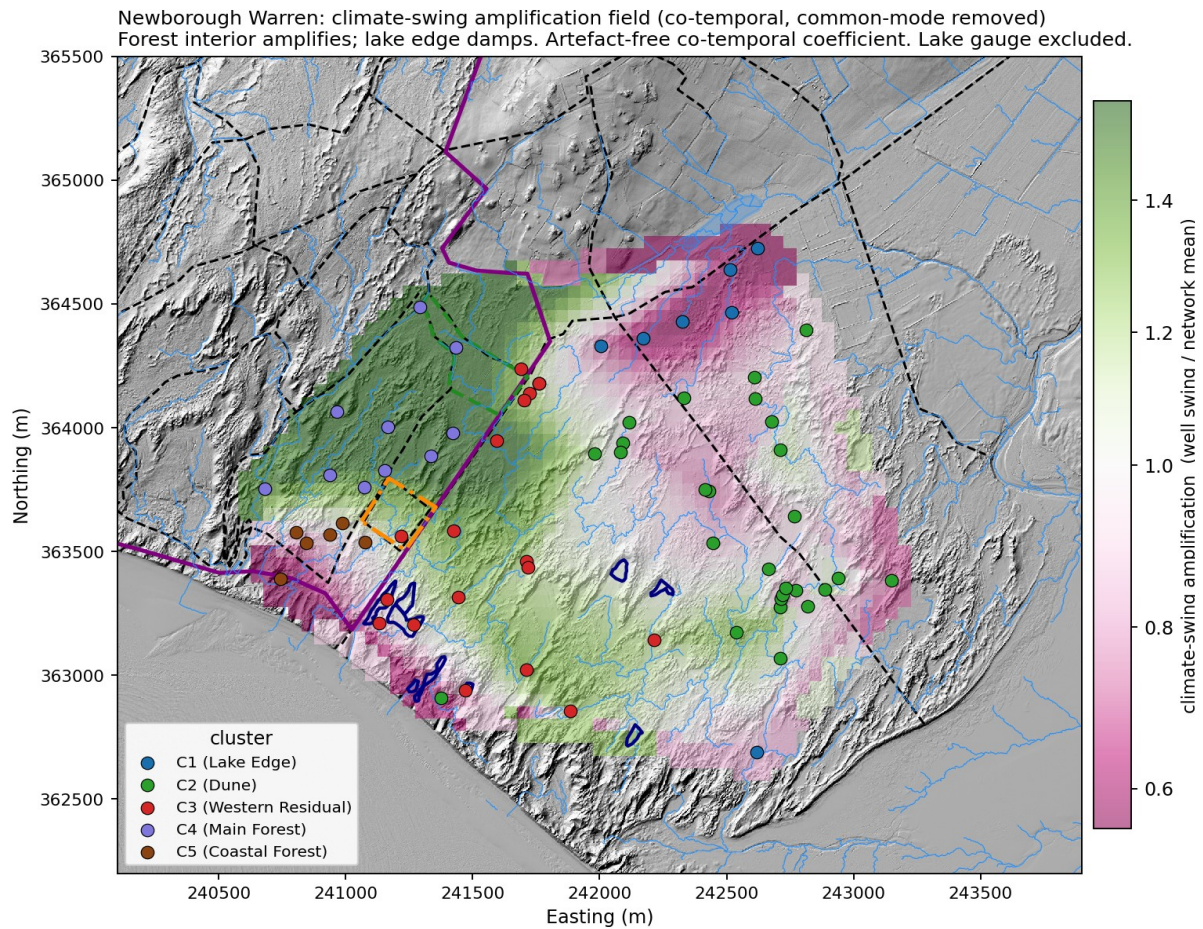


Figure 1.1.25: The climate-response envelope of the spring water table. Relative amplification field — each well's dry-to-wet spring swing as a ratio to the network-mean swing (≈ 0.73 m), with the common-mode removed; values above one amplify the shared climate swing, below one damp it. Sharp local steps between adjacent wells of unequal dry-year coverage — for example CEH9, which records all three dry springs, beside CEH39, which lacks the extreme 2012 spring — partly reflect that coverage difference rather than true structure and should be read with care. The Llyn Rhos-Ddu gauge excluded. (Source: 33_amplification_field.png).

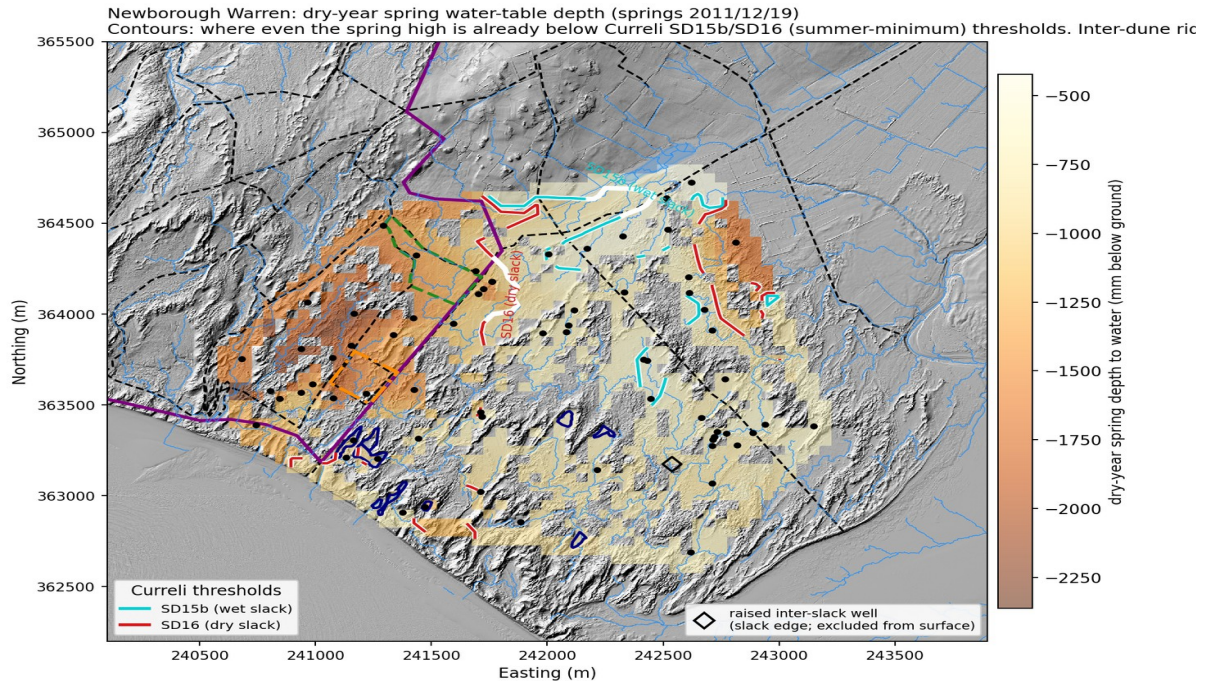


Figure 1.1.26: The climate-response envelope of the spring water table: Dry-year spring water-table depth — mean spring level across the dry extreme years (2011, 2012, 2019), with the Curreli et al. (2013) SD15b (0.61 m, wet slack) and SD16 (0.98 m, dry slack) thresholds contoured. CEH13, CEH14 and the Llyn Rhos-Ddu gauge excluded. The single raised inter-slack well (CEH10), sited on high ground between slacks and flooding only when the neighbouring slack overtops, registers the deep slack-edge drawdown directly; it is shown as a slack-edge marker but excluded from the slack-floor surface, which would otherwise smear its raised ground depth across the adjacent slacks. (Source: 33_dry_spring_depth.png).

Summary of pattern

Figure 60 predicts a large net loss across the forest interior under the parametric annual-mean drivers, and the 2017/2023 window comparison (Figure 62) shows a broadly deeper aquifer with the largest declines at the south-western coastal margin; but the absolute site-wide deepening is specific to that window pair (Section 5.7.5). Figure 63 confirms — with the common climate signal removed — that the coastal-margin decline is the dominant secular movement, and that the forest interior holds its position relative to the rest of the Warren. The per-cluster rates quantify this split: C4 Main Forest rises at a cluster mean of $+14.9 \text{ mm yr}^{-1}$ relative to the site (range $+8.4$ to $+20.5 \text{ mm yr}^{-1}$ across the nine reference wells; none individually significant after AR(1) correction); C1 Lake Edge declines at -8.0 mm yr^{-1} (CEH11 significant, $p = 0.027$); C5 Coastal Forest at -6.8 mm yr^{-1} (NW9 significant, $p = 0.039$); and C2 Dune is approximately neutral (-0.1 mm yr^{-1}). CEH22, at the south-western coastal margin, is the most rapidly declining well in the network at -26.5 mm yr^{-1} ($p < 0.001$). The site-mean spring trend over 2011–2025 is $+19.7 \text{ mm yr}^{-1}$ (AR-corrected $p = 0.275$; bootstrap CI -3.1 to $+42.8 \text{ mm yr}^{-1}$), reflecting the influence of the

exceptionally wet springs of 2021 and 2024 on a panel dominated by slowly draining wells; Figure 63 therefore shows the forest rising *relative to the network*, not in absolute recovery.

Figure 65 shows the forest amplifying the dry-to-wet climate swing more strongly than any other zone (C4 mean 1.72×, range 1.38–2.35×; C1 lake edge 0.61×; open dune near unity). The amplification correlates positively with β_2 ($r \approx +0.74$) and negatively with β_3 ($r = -0.45$, $p < 0.001$), consistent with slower-draining wells retaining wet-year recharge longer and therefore exhibiting larger spring swings — the same forest behaviour that inflates the C4 deepening in Figure 62, here measured directly rather than as an artefact of differencing.

The felled well FE2 (inside the December 2017 clearfell boundary) registers a spring differential trend of $+6.7 \text{ mm yr}^{-1}$ (Figure 63) and a co-temporal amplification of 1.23× (Figure 65). Both values lie notably below the unfelled C4 interior mean ($+14.9 \text{ mm yr}^{-1}$; 1.72×). FE2's wet-year spring depth (-1.06 m) rises less than the surrounding unfelled C4 reference wells (mean wet-year depth -0.56 m), and its swing (640 mm) is substantially smaller than the C4 mean (1,120 mm). This is physically consistent with canopy removal: the interception-dependent recharge suppression that drives the high wet-year amplification of the unfelled interior is absent at FE2. Concurrently, the summer drought floor at FE2 (-1.79 m in dry years) is slightly deeper than the C4 reference mean (-1.68 m), consistent with greater direct evaporation from exposed soil. FE1, approximately 20 m outside the felled boundary, shows an intermediate response ($+13.3 \text{ mm yr}^{-1}$; 1.53×). These observations are consistent with the clearfell interpretation of Section 4.6: felling attenuates the wet-year amplification that drives the apparent C4 differential rise, while not improving the summer minimum.

The reconciliation of the parametric prediction with the observed change — the window-sensitivity of the absolute deepening, the differential movement of Figure 63, and the climate amplification of Figure 65 — is taken up in Section 5.7.5.

Driver validation and comparative footing

The differential pattern above identifies where the network is moving, but not how much of that movement each modelled driver accounts for. Two diagnostics address this, and their shared conclusion is cautious: the network can bound the drivers' relative footing, but cannot independently confirm the coastal-retreat amplitude from the monitoring record alone.

A per-driver scale-factor regression (Section 3.8.5) fits the climate-corrected per-well change against the modelled coastal and clearfell fields as separate regressors, together with a free spatially-uniform intercept that absorbs the site-wide background decline present even at unfelled control wells. Across every window examined, the fitted scale factors are statistically indistinguishable from both zero and unity: the coastal factor is 0.53 (95% CI -0.12 to 1.18) over the full 2005–2025 record and 2.05 (-0.61 to 4.71) over the pre-intervention 2006–2012 window, while the clearfell factor over the post-clearfell 2018–2025

window is 3.62 (−0.77 to 8.00). The uniform intercept, by contrast, is consistently well-determined and large — a decline of 559 mm over 2006–2012 (95% CI −606 to −512) — indicating that the dominant secular movement in the network is common-mode rather than spatially structured by any single modelled driver. The regression explains only a modest fraction of the between-well variance ($R^2 \leq 0.30$ in every window).

This is a bounded null rather than a refutation. The coastal field cannot be independently confirmed here not because coastal influence is spatially narrow — the modelled reach ($L \approx 894$ m) extends across much of the western site — but because the signal cannot be statistically isolated: its spatial signature is collinear with easting ($r \approx -0.48$) and with the underlying clay-dip substrate gradient, the per-well residual scatter (of order ± 150 – 200 mm) exceeds the driver amplitudes being tested, and only around fifteen of the wells entering the fit carry a modelled coastal amplitude large enough to constrain the regression at all. The coastal shape (δ_0 , L) is moreover fitted from this same network via the gradient analysis of Section 4.9.4, so the scale-factor test is in part self-confirming; the one genuinely independent check is an expanding-window trajectory of the implied coastal decline rate, which is itself too contaminated by the 2006–2012 drought signal to be conclusive. The regression therefore neither confirms nor refutes the coastal-retreat interpretation of Section 4.9; it establishes that the present network lacks the spatial leverage to resolve it, and that background common-mode decline must not be attributed to any driver by default.

A complementary analysis (Section 3.8.5) places the drivers on a common footing over the 2005–2025 horizon, using observed BACI anchors where available (clearfell +119.6 mm; on-site scraping +129.4 mm; the reproducible WMC3 off-site scrape drawdown of −55 mm) and modelled fields elsewhere, together with a spatially-uniform climate / common-mode term ($c \approx -6.35$ mm yr^{−1}, ≈ -127 mm over the horizon) that carries the background warren-wide decline without attributing it to a specific mechanism. Expressed as a local water-table depth change, the coastal driver is the largest of the spatially-structured terms: a lowering of up to approximately 145 mm at the dune toe, tapering inland over a reach of roughly 900 m ($L \approx 894$ m); its depth-against-distance profile, alongside those of the forest, scraping and climate drivers, is shown in Figure 72. On the area-integrated volume and the ecological-threshold measures (Figure 68), however, the ranking inverts: the uniform climate / common-mode decline is the largest single term, because it acts across the whole site rather than a fringe, while the coastal drawdown — deep but confined to the western dune field — is second. Dune scraping produces a net site-wide loss because its off-site drainage cone draws down the water table across the area surrounding each cut by more than the on-site slack rise recovers — a scrape acts as a relief drain that draws water away rather than redistributing it into the slack, so the two do not balance; integrated across the site, the net scraping effect is of order 28% of the coastal drawdown in magnitude — a real secondary contributor to decline, but well short of the coastal term and not a primary driver in its own right. Sea-level rise, evaluated on the near-term five-year projection rather than the twenty-year horizon of the other terms, is indicative only and small by comparison. Both diagnostics

are first-order and do not close a water budget; the modelled coastal drawdown remains a projection anchored on the observed δ_0 and unconfirmed spatially by the network, while the off-site scrape drawdown is measured — reproducibly — at WMC3. Taken together they indicate coastal dominance at the western fringe on the local-depth measure, and dominance of the warren-wide climate / common-mode decline on the site-integrated footing, while placing an explicit bound on how far the monitoring record can substantiate the coastal amplitude itself. The expanding-window trajectory of the implied coastal decline rate — the one genuinely independent test of the coastal signal — is shown in Figure 67, and the comparative footing of all drivers in Figure 68.

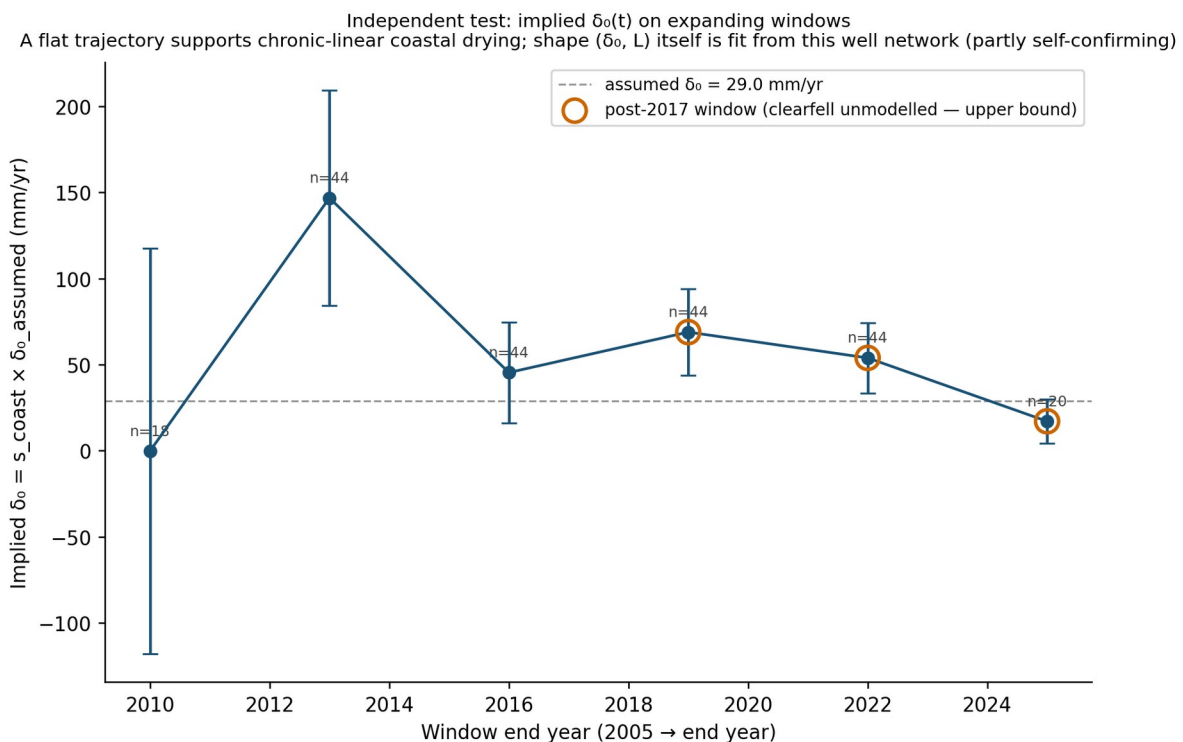
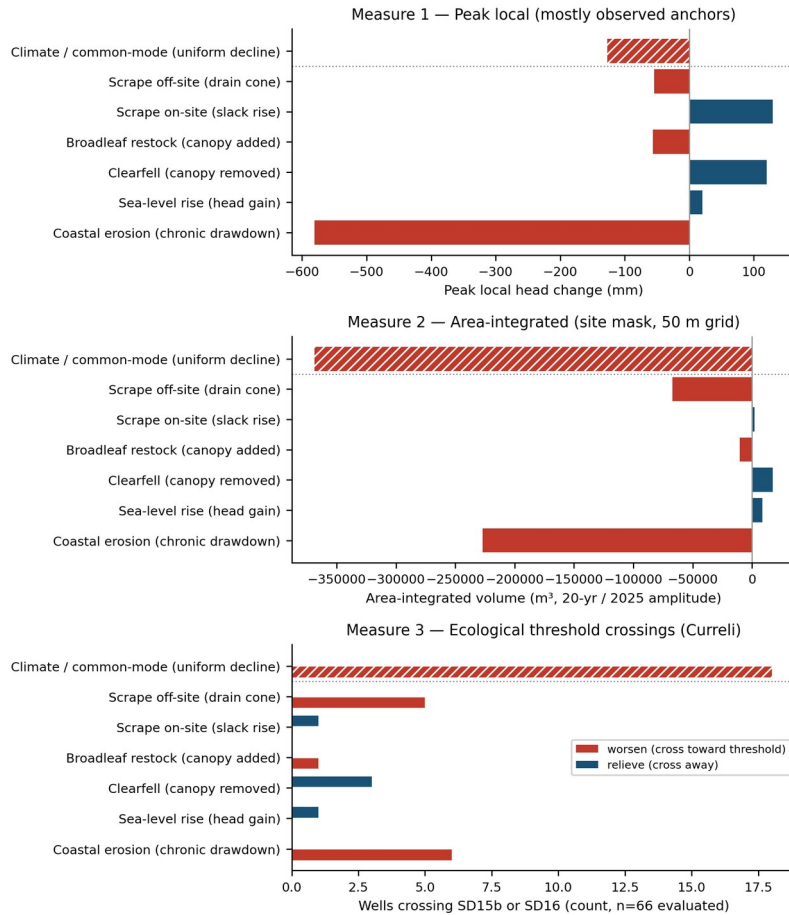


Figure 1.1.27: Independent test of the chronic coastal-drying assumption: implied coastal decline rate δ_0 from the coast-only scale-factor regression re-run on expanding windows (2005–2010 through 2005–2025). For each window the implied δ_0 is the fitted coastal scale factor multiplied by the assumed rate (29 mm yr⁻¹, dashed line). A trajectory that stays above zero throughout is the signature a steadily propagating erosional drawdown would produce — the coast-to-inland gradient steepening year on year; the value relative to the 29 mm yr⁻¹ line then indicates whether that drawdown is proceeding at, above, or below the assumed chronic rate. A dip to zero indicates no detectable coastal signal on this line (a flat head difference, consistent with static substrate geometry rather than a moving erosional boundary), and a value below zero would indicate a reversing gradient that refutes chronic coastal drawdown — though in the short early windows a near-zero value more likely reflects insufficient statistical power to resolve the signal than a genuine absence of it. The three fully pre-clearfell windows (ending 2010, 2013, 2016) are the only clean coast-only reads but are individually too noisy — the earliest is dominated by the 2006–2012 drought — to establish a stable rate; the windows ending 2019 or later, ringed in orange, span the December 2017 clearfell without a term to absorb its recovery signal, so their implied δ_0 is an upper bound on the coastal contribution rather than a direct estimate. The test is therefore inconclusive: it neither confirms nor refutes the modelled coastal amplitude, consistent with the bounded null reported in the text. (Source: 37_implied_delta0_trajectory.png)

Part B — comparative driver footing v1.2.0
forest · scrape · coast · climate on common measures (2005→2025)



Climate / common-mode (hatched, below the separator) is the spatially-uniform background decline — shown apart from the mechanism-specific drivers and not attributed to a single driver.

Figure 1.1.28: Comparative footing of the drivers over the 2005–2025 horizon, on three common measures — peak local head change (top, mm), area-integrated volume (middle, m³; the head field integrated over each footprint with a representative specific yield), and ecological threshold crossings (bottom; wells moved across the Curreli et al. (2013) SD15b wet-slack and SD16 dry-slack thresholds). Amplitudes are the observed BACI anchors (clearfell, on-site scraping, and the reproducible WMC3 off-site drawdown), the modelled Script 20 fields (coastal drawdown and sea-level rise, the wider off-site scrape cone, broadleaf restock), and the modelled spatially-uniform climate / common-mode term ($c \approx -6.35 \text{ mm yr}^{-1}$, -127 mm over the horizon). Coastal drawdown is modelled and spatially unconfirmed by the network; the off-site scrape drain is measured near-field at WMC3 — reproducibly, across the 2015 and 2023 scrapes — with only its wider cone modelled (§4.9.8). The climate / common-mode term is shown apart from the spatially-structured drivers (hatched, below the separator) as the background decline not attributed to a single mechanism. Fields are superposed linearly, an upper bound in overlap zones that does not close a water budget. (Source: 37b_driver_footing.png)

1.3 Scenario Analysis

The per-well SSM coefficients derived in Section 4.9 were used to project equilibrium water table responses under prescribed climate and forest management perturbations, following the framework of Section 3.8. Six scenarios were evaluated: baseline; UKCP18 2050s and 2080s climate trajectories; full clearfell; 50% thinning; and broadleaf conversion. Scenario difference maps were computed as $\Delta h = h(\text{scenario}) - h(\text{baseline})$ for each well, season and cluster. The full per-scenario, per-season, per-cluster Δh output is available through the Newborough Warren Hydrogeological Modelling Suite (Hollingham, 2026b), an interactive web-based scenario viewer that renders the per-well Δh field across the site using IDW interpolation, with a complementary depth-below-surface view masked to hydrologically meaningful slack positions and overlaid with the Curreli et al. (2013) SD15b/SD16 ecological thresholds. The cluster-level results are summarized below.

1.3.1 Climate Scenario Projections

Climate scenario responses are spatially coherent and physically consistent with the per-cluster β coefficient structure. Under UKCP18 2050s forcing, the site-wide mean annual Δh is -0.018 m/month, with cluster-level annual responses ranging from -0.026 m/month at C4 (Main Forest) to -0.009 m/month at C1 (Lake Edge). Under UKCP18 2080s forcing, the site-wide mean annual response deepens to -0.033 m/month, with a maximum of -0.047 m/month at C4 and a minimum of -0.017 m/month at C1. Both UKCP18 scenarios show the forested clusters carrying the largest annual declines, driven by the combination of high β_2 (atmospheric draw sensitivity) and reduced net recharge under interception. The annual response conceals a pronounced seasonal asymmetry. Under both trajectories, winter Δh is positive at all clusters (site-wide mean $+0.028$ m/month at 2050s and $+0.056$ m/month at 2080s) as the increased winter rainfall under RCP8.5 Wales outpaces the concurrent increase in winter PET. Summer Δh is strongly negative and carries the annual signal: site-wide means of -0.064 m/month at 2050s and -0.121 m/month at 2080s, reaching -0.134 m/month at C2 under the 2080s trajectory. This summer-dominated signature reflects the same physical mechanism established in the observed record (Section 4.8.2): at Newborough, hydrological deterioration under climate change operates through intensified summer atmospheric demand rather than declining winter recharge.

The same projection framework, applied to the five-year mean spring water level (MSL5; Section 4.8.4), gives a markedly smaller response than the summer-minimum metric. The projected MSL5 shift is -12 to -21 mm across the clusters at the 2050s horizon and -21 to -39 mm at the 2080s horizon (Figure 69), against summer Δh of -38 to -70 mm and -71 to -134 mm respectively over the same horizons. The MSL5 response is three to five times smaller than the summer response in every cluster and at both horizons. The contrast is mechanistic, not incidental: MSL5 is computed over the March–May spring window, which receives the projected increase in winter–spring rainfall, partly offsetting the concurrent increase in spring evaporative demand; the summer minimum receives the full projected

increase in summer atmospheric demand compounded by the projected reduction in summer rainfall. The spring baseline metric is therefore comparatively insulated against the projected trajectory, while the summer-minimum metric is highly exposed. Expressed against the Curreli et al. (2013) thresholds, the projected MSL5 shifts do not move any cluster across a threshold under either horizon — C1 and C2 remain within the SD15b wet-slack regime, C4 remains in the SD16 dry-slack regime, and C5 remains below SD16 — whereas the parallel summer-minimum deterioration substantially worsens threshold exceedance in the open-dune clusters whose summer minima already approach the dry-slack threshold (Section 4.8.1). The implication for the binding ecological constraint is taken up in Section 5.2.5.

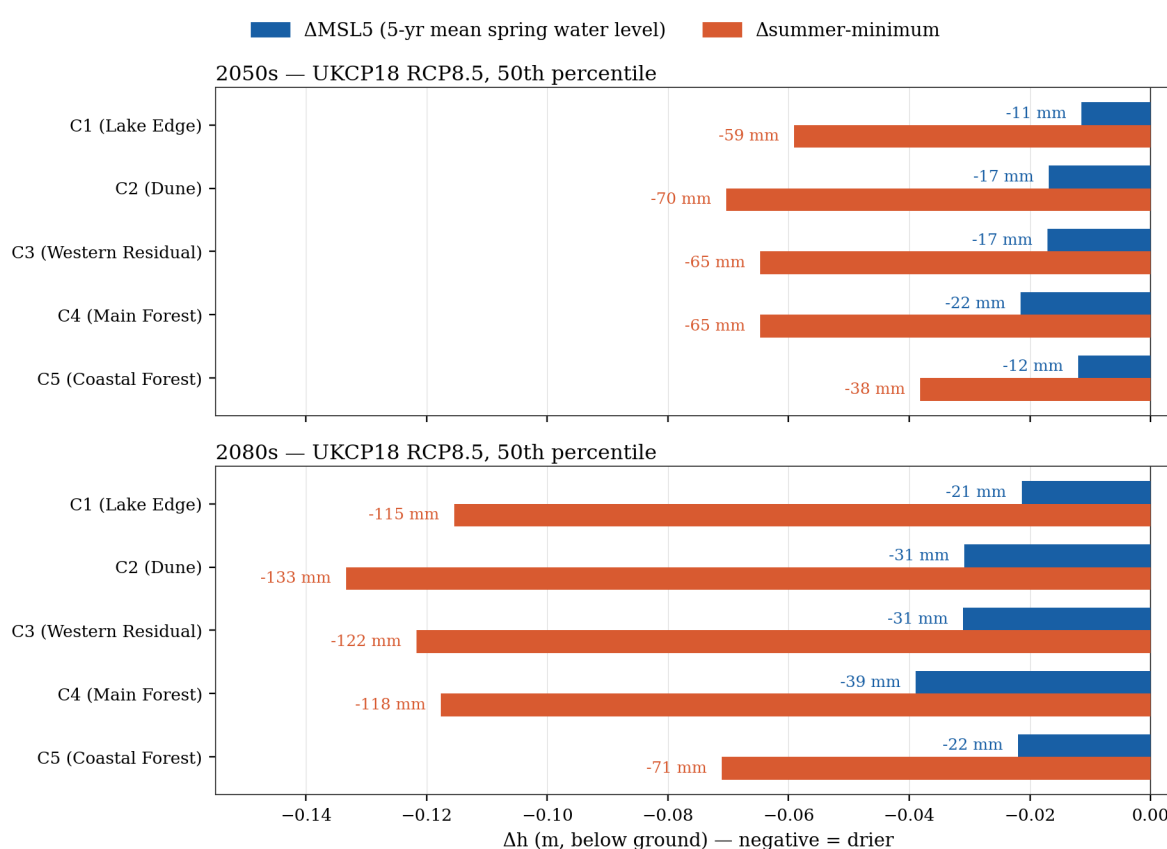


Figure 1.1.29: Projected shift in the five-year mean spring water level (ΔMSL5 , blue) compared with the projected shift in summer-minimum water level ($\Delta\text{summer-minimum}$, orange), by cluster, under UKCP18 RCP8.5 forcing: 2050s (upper panel) and 2080s (lower panel). All shifts are negative (deeper, drier); bar length is the magnitude of the projected deepening in millimetres. In every cluster and at both horizons the summer-minimum response is three to five times the MSL5 response — the spring baseline metric is comparatively insulated against the projected trajectory while the summer minimum is highly exposed (Section 4.10.1). The contrast is seasonal in origin: the March–May window over which MSL5 is computed receives the projected increase in winter–spring rainfall, partly offsetting the rise in spring evaporative demand, whereas the summer minimum receives the full projected increase in summer atmospheric demand compounded by the projected reduction in summer rainfall. ΔMSL5 values are the cluster-centroid projections from Script 26b; $\Delta\text{summer-minimum}$ values are the seasonal-mean Δh from the Script 19 scenario framework (Section 4.10.1). (Source: [fig_msl5_vs_summer_min_projection.png](#))

1.3.2 Forest Management Scenarios

Forest management scenarios produce responses at the two forested clusters (C4 and C5), with no effect at C1–C3. Full clearfell is modelled by setting canopy interception to 0% and adjusting β_2 at C4 and C5 wells. Fifty percent thinning halves interception from 24% to 12% with proportional β_2 adjustment. Broadleaf conversion replaces the Corsican pine canopy interception fraction of 24% (Freeman, 2008) with an annual-mean deciduous interception of 15% (Komatsu et al., 2011), and applies a seasonally varying β_2 adjustment to reflect deciduous phenology: β_2 follows the winter and summer broadleaf coefficients derived in Section 3.8, held near baseline during the leafless dormant season (November–March) and raised during the growing season to reflect the comparable transpiration rates of mature deciduous and Corsican pine canopy in full leaf.

Under full clearfell with the BACI-corrected β_2 multiplier (Section 3.8), the framework predicts a modest annual water table rise at both forest clusters (C4: +0.039 m/month head, equivalent to +8.1 mm w.e./month volumetric; C5: +0.039 m/month head, +10.4 mm w.e./month). The annual response is driven primarily by the winter recharge gain from canopy removal (C4: +0.047 m/month head; C5: +0.045 m/month head). The summer response is positive but smaller at both clusters (C4: +0.031 m/month; C5: +0.032 m/month), reflecting that the winter interception gain exceeds the summer β_2 cost across the year at both substrates. Fifty percent thinning produces a similar pattern at reduced magnitude (C4: +0.020 m/month annually, +4.0 mm w.e./month; C5: +0.019 m/month, +5.2 mm w.e./month). These are steady-state predictions from the SSM coefficient structure; the predicted direction is consistent with the BACI monitoring record, which detects a significant mean monthly recovery at the Impact tier (+120 mm against the Forest control; Section 4.6.3) but no corresponding improvement in summer minima (Section 4.6.4; further interpreted in Section 5.5).

Broadleaf conversion produces a near-neutral annual response at both forest clusters (C4: +0.011 m/month head; C5: +0.013 m/month head), masking a pronounced seasonal asymmetry: winter head-space values are among the largest of any forestry scenario (C4: +0.026 m/month; C5: +0.021 m/month) reflecting the near-zero interception under leafless canopy during the recharge season, but the summer response is slightly negative at C4 (−0.003 m/month) and near-zero at C5 (+0.005 m/month) as the growing-season transpiration penalty under mature deciduous canopy consumes the interception gain. The net annual volumetric effect is −0.7 mm w.e./month at C4 and +1.6 mm w.e./month at C5. The additional winter recharge steepens the local hydraulic gradient and drains away through the highly transmissive sand before the summer minimum is reached (further interpreted in Section 5.5.2).

Site-wide means under forest management are small (annual +0.008 m/month clearfell, +0.004 m/month thinning) because only 14 of 66 reference wells are in the two forest

clusters and the framework applies no spatial propagation from the forest zone into other clusters. Broadleaf conversion produces no meaningful site-wide signal.